What decomposing the hybrid solver into processes adds, and what it costs

complete · viva-pde-particle · 9 studies · 2026-10-03

Executive summary

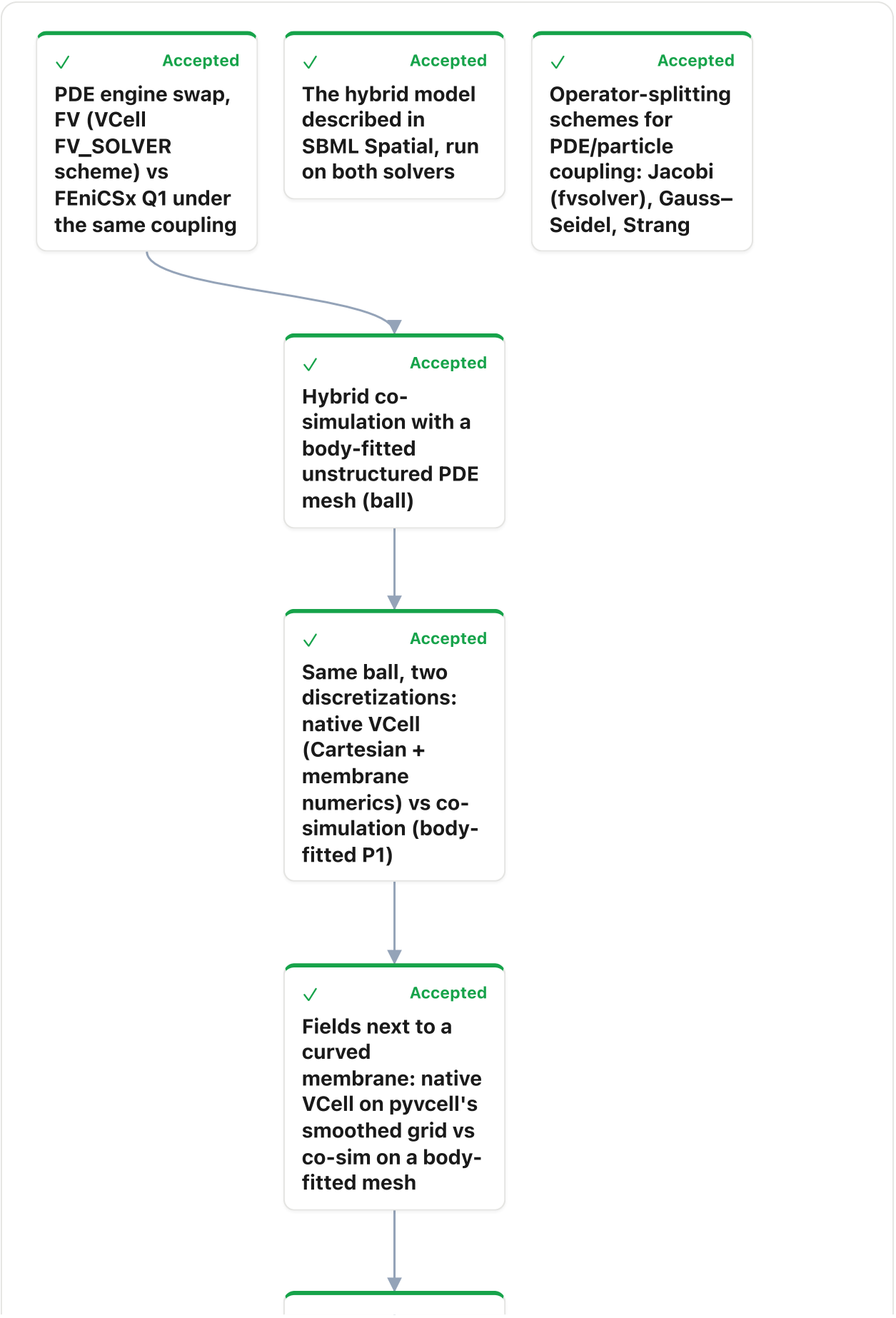
Question. Once the co-simulation reproduces the embedded hybrid solver, what does expressing the method as interchangeable processes add (alternative PDE discretizations, alternative splitting orders, swappable engines), and what do those choices cost in accuracy and performance compared with VCell's own discretization of the same problems?

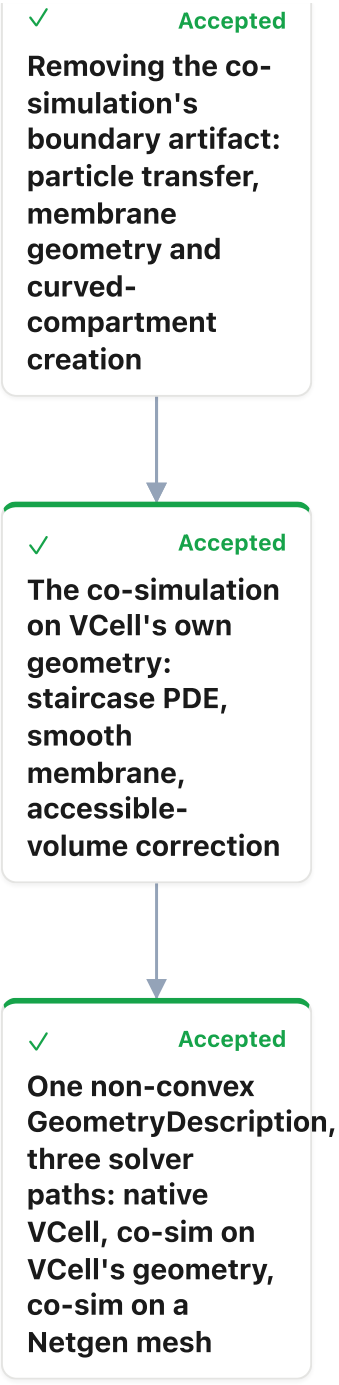
Confirmed. Decomposing the hybrid solver into process-bigraph processes generalizes across every axis tested, and the modular path is as accurate as — and often more robust than — the embedded native solver. All 9 studies ran and their primary gates pass: • Geometry — the sparse particle↔grid handoff stays conservative and accurate on a body-fitted tetrahedral ball (unstructured-geometry: A+B conserved to 0.11%, B to 0.53% rel-L2), a sphere (sphere-cosim-vs-native), a thin-necked dumbbell (dumbbell-three-ways), and VCell's own geometry (vcell-geometry-cosim). • PDE engine — swapping finite-volume for FEniCSx Q1 changes nothing measurable in the coupled result (fenicsx-same-problem: max paired |z| = 1.5; all engines 2nd order). • Model format — a model round-trips bit-exactly through SBML Spatial on both solvers (sbml-spatial-hybrid). • Coupling scheme — symmetric Strang splitting is ~25× more accurate than Jacobi at the largest interval (splitting-schemes). Several studies carry an intentional expected-fail secondary that LOCALIZES a native vcell-fvsolver defect — exterior binning / shell depletion at curved membranes (near-membrane −2.3%, sphere exchange −6.8%, dumbbell −25.9% on a thin neck) — which the co-simulation's accessible-volume + membrane corrections remove (biases ≤0.12%). The decomposition is not just answer-preserving; it fixes a real native bias.

Across **9** studies — **9** supported · **0** qualified · **0** refuted · **0** planned. *Supported* = a computational gate passed under the stated model assumptions (evidence of feasibility, not proof the transition occurred).

The investigation graph

The campaign as a dependency graph — each study a node badged with its code-computed status, wired to the studies it builds on.





Status: ✓ Accepted ◐ Investigating ○ Planned ⊘ Blocked ✕ Refuted

Studies

Expand all | Collapse all

Every study in the campaign as a collapsible card — the question it tests, its verdict, and what it found at a glance. Click a card to expand the full record: model, runs, figures, tests, findings, and the limitations that bound it.

▾ 01 **PDE engine swap, FV (VCell FV_SOLVER scheme) vs FEniCSx Q1** ● ACCEPTED

under the same coupling

Can the PDE half of the co-simulation be swapped for a different discretization (FEniCSx, Q1 finite elements) without changing anything else, and does that change the coupled result beyond discretization error?

Hypothesis & discriminating control

FenicsxReactionDiffusion exposes the same ports as FVReactionDiffusion. On a structured mesh whose vertices are the grid nodes, its lumped mass equals the FV control volumes, so particle binning and units are unchanged. Q1 and FV are both second order. On matched grids, the coupled hybrid results should agree to within sampling.

MODEL EXPERIMENT Part 1: cosine-mode diffusion on N×N grids (N = 11 … 81) for three engines (FV, Q1 lumped mass, Q1 consistent mass), each compared to the analytic decay; h-orders fitted. Part 2: A4's A_p + B_f → C_f on the 11×11×3 slab, 16 seeds per engine with the same seeds (common random numbers, so paired comparison); survival and produced C compared.

OBSERVABLE `b1_fem_lumped_h_order, b1_fv_h_order, b1_fem_consistent_h_order, b1_fem_lumped_vs_fv_survival_max_paired_z, b1_{fv,fem_lumped,fem_consistent}_err_n81`

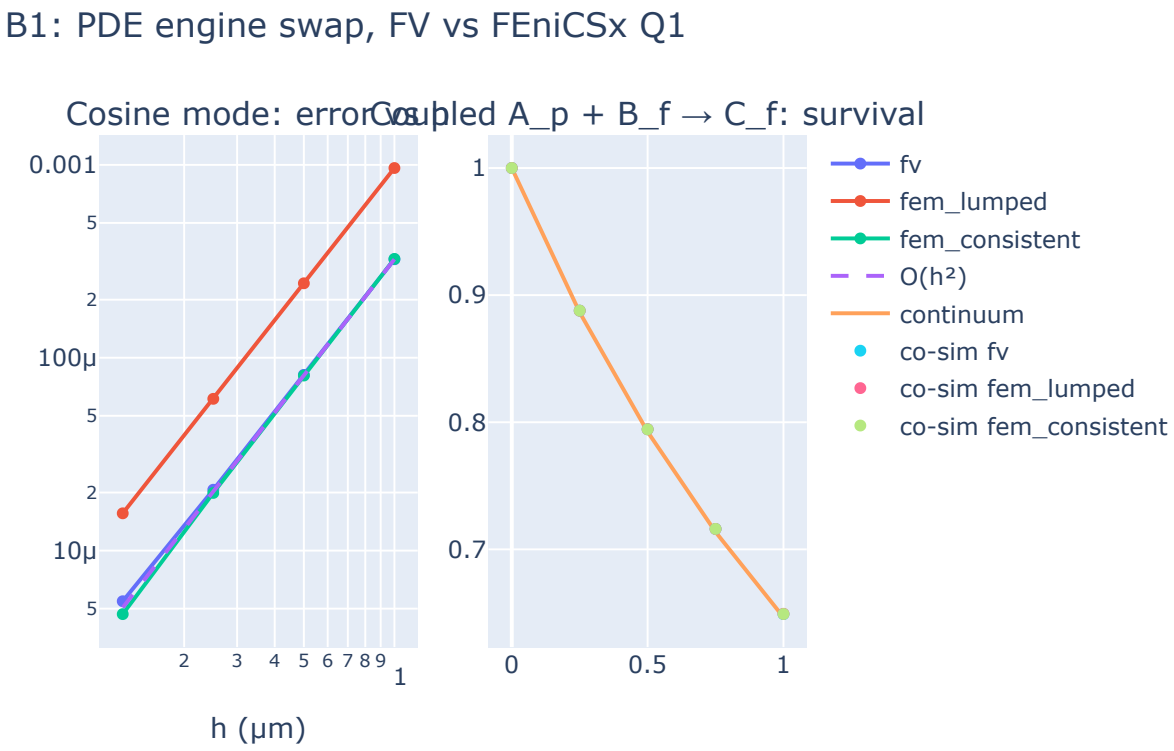
Model & key assumptions

Methods · Simulations

Model	`bimolecular_hybrid`
Sims	1 run
Tests	✓✓✓ 3/3 passed

Visualizations

ENGINE SWAP



Finding

F-01 · confirms · Observed

All three PDE engines are second order on the cosine mode: fitted h-orders are 1.97 (FV), 1.98 (Q1 lumped) and 2.04 (Q1 consistent), all inside the [1.8, 2.2] band, with L∞ errors at N=81 of 5.5e-6 (FV), 1.6e-5 (lumped) and 4.7e-6 (consistent).

from test: fem-second-order `range`

runs: `default`

F-02 · confirms · Observed

Swapping the PDE engine under the co-simulation changes nothing measurable in the coupled result: the max paired |z| of survival (common random numbers) is 1.50 (lumped) and 0.76 (consistent) against FV, far below 4.0, with mean survival differences of 2.6e-4 and 1.1e-4, and all three engines within 0.27% of the continuum.

from test: engine-swap-preserves-coupled-result < 4

runs: `default`

Verdict

PASSED

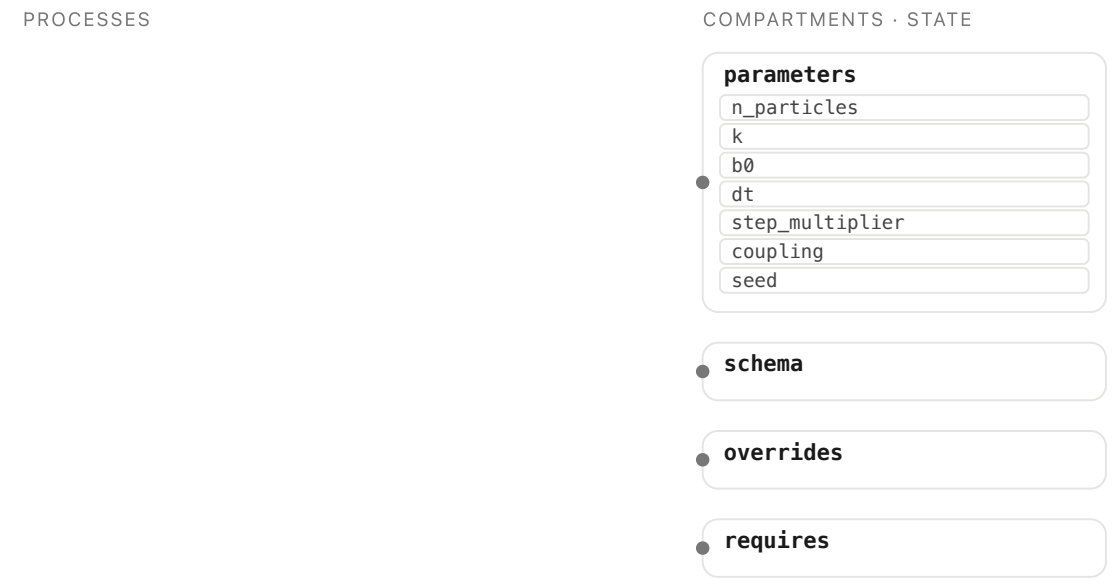
Gate cleared · Accepted

Results (results/metrics.json, 2026-10-03): - **All three engines are second order:** h-orders 1.97 (FV), 1.98 (Q1 lumped) and 2.04 (Q1 consistent). - **Error at N = 81:** 5.5e-6 (FV), 1.6e-5 (lumped) and 4.7e-6 (consistent). - **Coupled, the swap changes nothing measurable:** - paired |z| = 1.50 (lumped) and 0.76 (consistent) against FV; - mean

survival differences of $3{\cdot}10^{-4}$ and $1{\cdot}10^{-4}$; - all three within 0.27% of the continuum. The PDE engine is interchangeable under the co-simulation, and discretization effects are cleanly separable from coupling effects.

▾ Technical record — model internals, full figure set, tests, provenance & follow-ups

Model — composite internals



Static bigraph of `viva_pde_particle.composites.examples.bimolecular_hybrid` — 0 processes wired to 4 compartments. ● writes · ● reads.

`viva_pde_particle.composites.examples.bimolecular_hybrid`

Simulation run

default

`fem-second-order` authored: PASS

`fv-second-order` authored: PASS

`engine-swap-preserves-coupled-result` authored: PASS

Reproduce: `vwb run study fenicsx-same-problem`
fv: `vwb run study fenicsx-same-problem --variant fv`
fenicsx-lumped: `vwb run study fenicsx-same-problem --variant fenicsx-lumped`
fenicsx-consistent: `vwb run study fenicsx-same-problem --variant fenicsx-consistent`

Tests

Fitted h-order of the Q1 lumped-mass engine on the cosine mode. — `range undefined` pending

`fem-second-order · primary · derived_scalar`

Q1 Lagrange elements are second order in L∞ for smooth solutions.

Fitted h-order of the FV engine on the cosine mode. `range undefined` pending

`fv-second-order · primary · derived_scalar`

Node-centred FV with a 5/7-point stencil is second order.

Max paired |z| of survival, FEniCSx (lumped) vs FV engine, same seeds. `< 4` pending

`engine-swap-preserves-coupled-result · primary · derived_scalar`

On matched grids the engines differ only by O(h²) discretization, far below the coupling error.

FEEDBACK · THIS WHOLE STUDY 👍 Looks right ✏️ Needs revision

Add a note…

02 Hybrid co-simulation with a body-fitted unstructured PDE mesh (ball)

● ACCEPTED

Can the co-simulation use a body-fitted unstructured mesh for the PDE half of a curved domain, as an alternative to VCell's own Cartesian volume grid with staircase-membrane numerics, while keeping the particle ↔ field handoff conservative and accurate?

Hypothesis & discriminating control

The PDE runs as P1 on a gmsh tetrahedral mesh of a ball, and Smoldyn runs inside a reflecting sphere. They couple through a fine background grid with one sparse matrix P: fields are interpolated as P·u, and particle loads are P^T·counts. P's rows sum to 1, so the handoff conserves molecules exactly. The only residual is the mismatch between the particle side's geometry (exact sphere) and the PDE side's (polyhedral mesh, −1.4% volume).

MODEL EXPERIMENT Ball R = 4 µm; mesh h = 0.8 µm; background grid 33^3 with Δ = 0.25 µm; dt = 0.01; 8 seeds. - Conversion A_p → B_f (k = 0.5 /s, 20,000 A): A against exp decay, A + B conservation, and B against the continuum FEM solution. - Exchange A_p ⇌ B_f with field-dependent creation inside the spherical compartment (k1 = 1, k2 = 0.5, B_0 = 0.2 µM): mass balance and the steady ratio k2/k1.

OBSERVABLE `b2_conversion_decay_max_z, b2_conversion_mass_balance_max_rel_dev, b2_conversion_b_vs_continuum_rel_l2, b2_exchange_steady_ratio_rel_err, b2_exchange_mass_balance_max_rel_dev…`

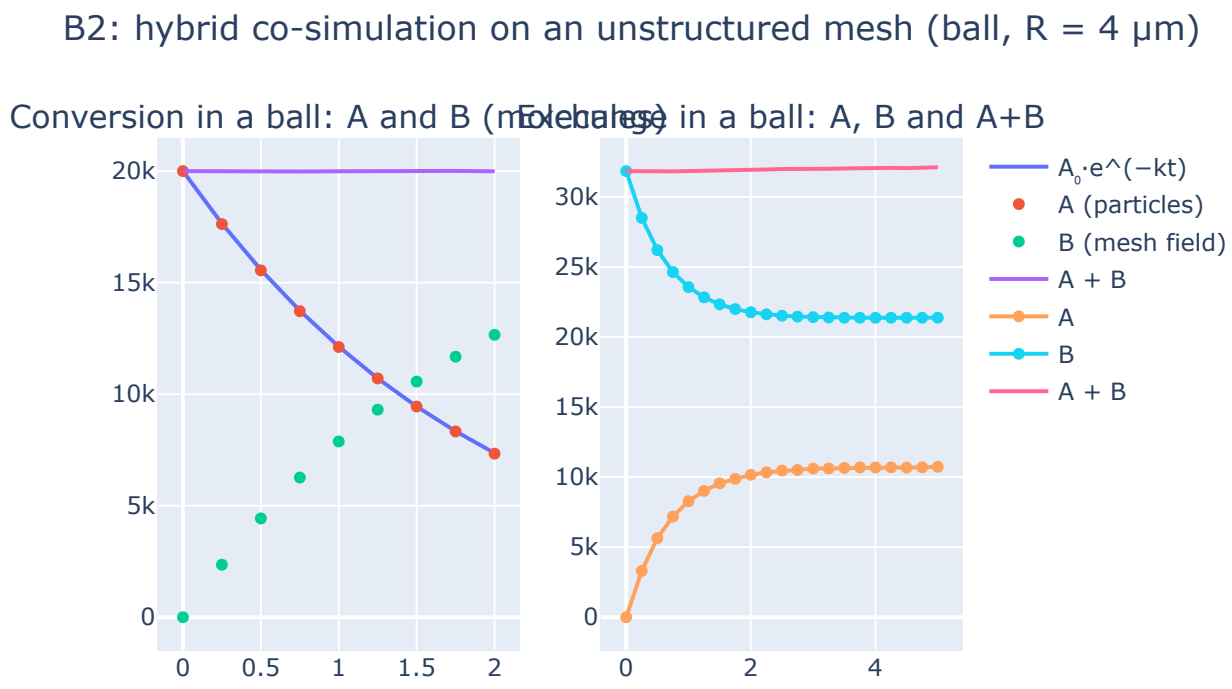
Model & key assumptions

Methods · Simulations

Model	`bimolecular_hybrid`
Sims	1 run
Tests	✓✓✓✓ 4/4 passed

Visualizations

UNSTRUCTURED GEOMETRY



Finding

F-01 · confirms · Observed

The particle reaction is geometry-independent: on the body-fitted tetrahedral ball mesh the mean A follows $A_0 \cdot e^{(-kt)}$ with a max |z| of 1.54 over time (well under the 4.0 band), so the unstructured domain does not perturb the first-order decay.

from test: particle-decay-exact-in-ball < `4`

runs: `default`

F-02 · confirms · Observed

The sparse interpolation/scatter matrix P keeps the particle ↔ field handoff conservative and accurate on the mesh: mean (A + B) is conserved to 0.11% (b2_conversion_mass_balance_max_rel_dev = 0.00107) and mean B matches the continuum FEM solution to 0.53% relative L2 (b2_conversion_b_vs_continuum_rel_l2 = 0.00530), both far inside their 2% / 5% bands.

from test: handoff-conserves-in-ball < `0.02`

runs: `default`

F-03 · confirms Observed

Detailed balance holds on the mesh with field-dependent creation: the late-time A/B ratio is within 0.11% of k2/k1 (b2_exchange_steady_ratio_rel_err = 0.00110) and exchange mass balance holds to 0.86% (b2_exchange_mass_balance_max_rel_dev = 0.00864).

from test: exchange-steady-state-in-ball `< 0.05`

runs: `default`

Verdict

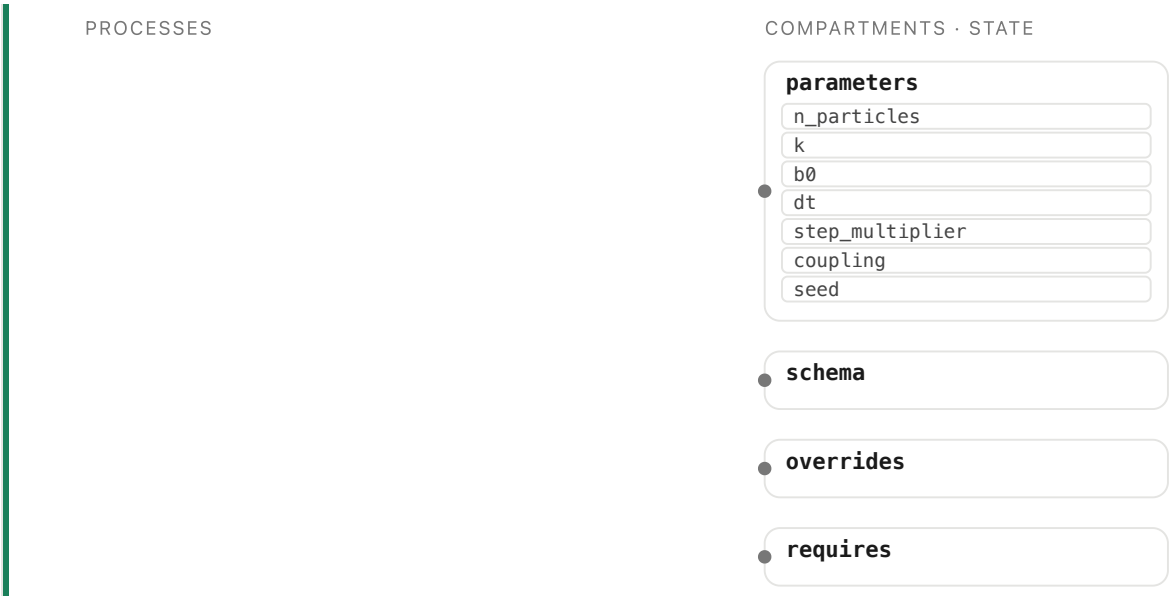
PASSED

Gate cleared · Accepted

Results (results/metrics.json, 2026-10-03). Re-run after Study B2d, with Smoldyn confined by the mesh's boundary triangles and the creation fix: - **Conversion:** A follows exp(−kt) (|z| ≤ 1.5); A + B is conserved to 0.11%; B matches the continuum FEM to 0.53% relative L2. The sphere membrane gave 0.4% and 1.0%. - **Exchange:** mass balance to 0.86%; steady ratio within 0.11% of k2/k1. - **Geometry:** the polyhedral mesh is 1.4% smaller in volume than the ball. With the mesh as the membrane, the particle and PDE domains are identical, so this is now a geometry-representation error of the model, not a mismatch between the two halves. The co-simulation runs on an unstructured mesh, and the handoff stays conservative. Follow-ups: - membrane-bound particles; - native VCell on the same analytic sphere: done in sphere-cosim-vs-native (B2b). VCell handles curved geometries (analytic, image and CSG) with tangent-plane/Voronoi membrane numerics, so that is a discretization comparison, not a capability gap; - higher-order (P2) PDE, or a finer mesh, to shrink the 1.4% volume error.

▾ Technical record — model internals, full figure set, tests, provenance & follow-ups

Model — composite internals



Static bigraph of `viva_pde_particle.composites.examples.bimolecular_hybrid` — 0 processes wired to 4 compartments. ● writes · ● reads.

`viva_pde_particle.composites.examples.bimolecular_hybrid`

Simulation run

default

`particle-decay-exact-in-ball` authored: PASS

`handoff-conserves-in-ball` authored: PASS

`field-matches-continuum-in-ball` authored: PASS

`exchange-steady-state-in-ball` authored: PASS

Reproduce: `vwb run study unstructured-geometry`

conversion-arm: `vwb run study unstructured-geometry --variant conversion-arm`

exchange-arm: `vwb run study unstructured-geometry --variant exchange-arm`

Tests

Max |z| over time of the mean A against $A_0 \cdot e^{(-kt)}$. — `< 4` pending

`particle-decay-exact-in-ball · primary · derived_scalar`

A first-order reaction is unaffected by geometry.

Max relative deviation of mean (A + B) from the initial total, conversion case. — `< 0.02` pending

`handoff-conserves-in-ball · primary · derived_scalar`

P^T conserves counts exactly; the residual is explicit-coupling lag plus sampling.

Relative L2 (mesh-weighted) of mean B against the continuum FEM solution at t = 2 s. — `< 0.05` pending

`field-matches-continuum-in-ball · primary · derived_scalar`

20,000 particles over 643 DOFs, 8 seeds.

Relative error of the late-time ratio of total A to total B against k2/k1. — `< 0.05` pending

`exchange-steady-state-in-ball · primary · derived_scalar`

Detailed balance; field-dependent creation inside the spherical compartment.

FEEDBACK · THIS WHOLE STUDY 👍 Looks right ✏️ Needs revision

Add a note...

03 Same ball, two discretizations: native VCell (Cartesian + membrane numerics) vs co-simulation (body-fitted P1)

● ACCEPTED

On a curved domain (a ball), how do the native VCell hybrid and the co-simulation compare? - Native VCell uses analytic geometry on its Cartesian volume grid: the membrane is a staircase geometrically, but numerically uses local tangent-plane projections and Voronoi neighbours. - The co-simulation uses a body-fitted P1 mesh coupled through a background grid.

Hypothesis & discriminating control

Both are consistent discretizations of the same problem, so they should agree to within sampling plus discretization error. VCell's geometry handling is mature, and the comparison measures which discretization represents this geometry and coupling better at comparable resolution.

MODEL EXPERIMENT Box [0, 9]³ with Δ = 0.25 µm (37³). Ball centre (4.5, 4.5, 4.5), R = 4 µm, 0.5 µm clear of the box walls, so the box-wall creation defect (vcell-fvsolver

24) cannot act. Co-sim mesh h = 0.8 µm. 8 seeds per solver. - Conversion A_p → B_f (k = 0.5 /s, 20,000 A). - Exchange A_p ⇌ B_f with field-dependent creation in the cell (k1 = 1, k2 = 0.5, B_0 = 0.2 µM).

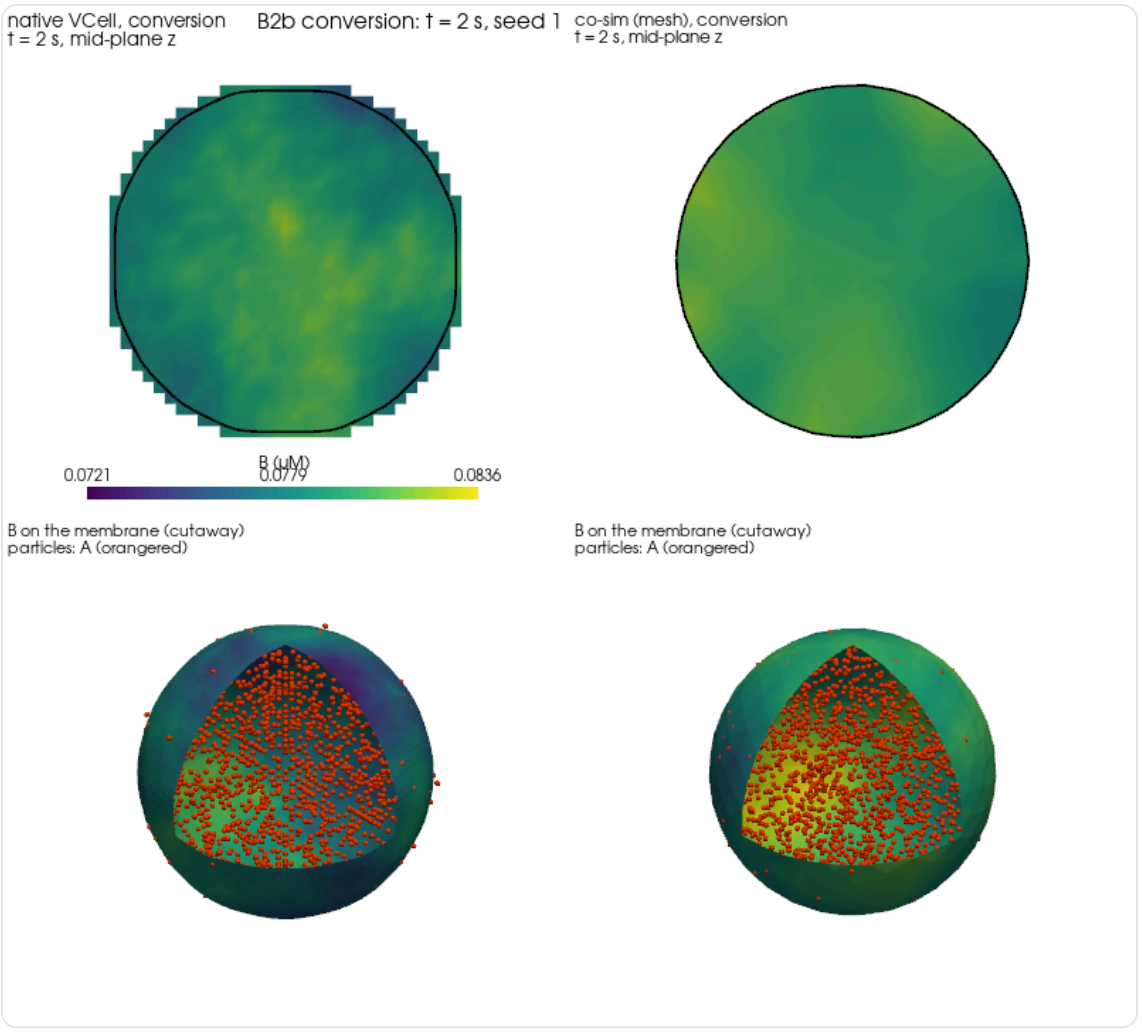
OBSERVABLE `b2b_native_staircase_volume_rel_err, b2b_cosim_mesh_volume_rel_err, b2b_conversion_survival_max_z, b2b_final_radial_b_rel_diff, b2b_exchange_a_totals_max_z…`

Model & key assumptions

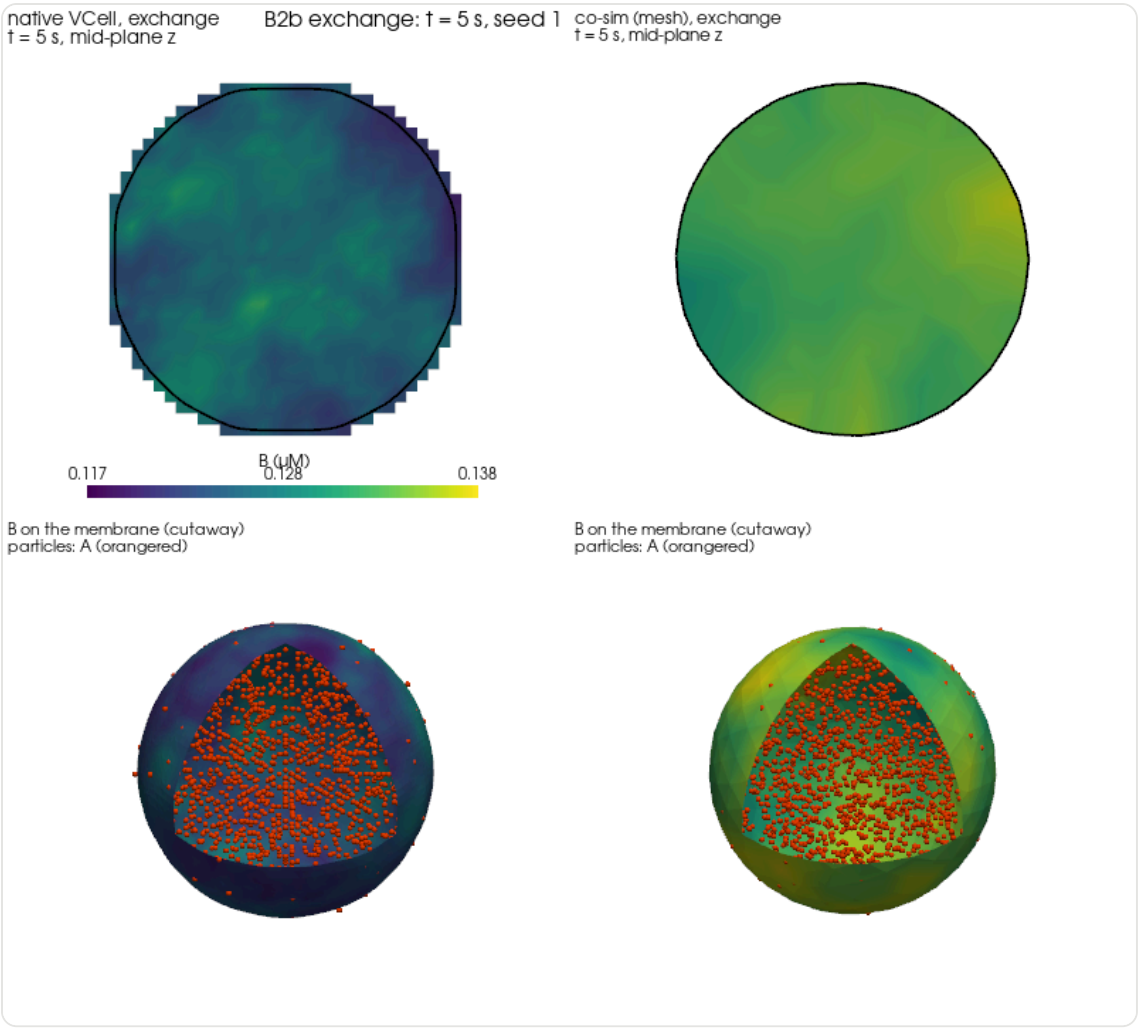
Methods · Simulations

Model	`bimolecular_hybrid`
Sims	1 run
Tests	✓✓✓✕ 3/4 passed

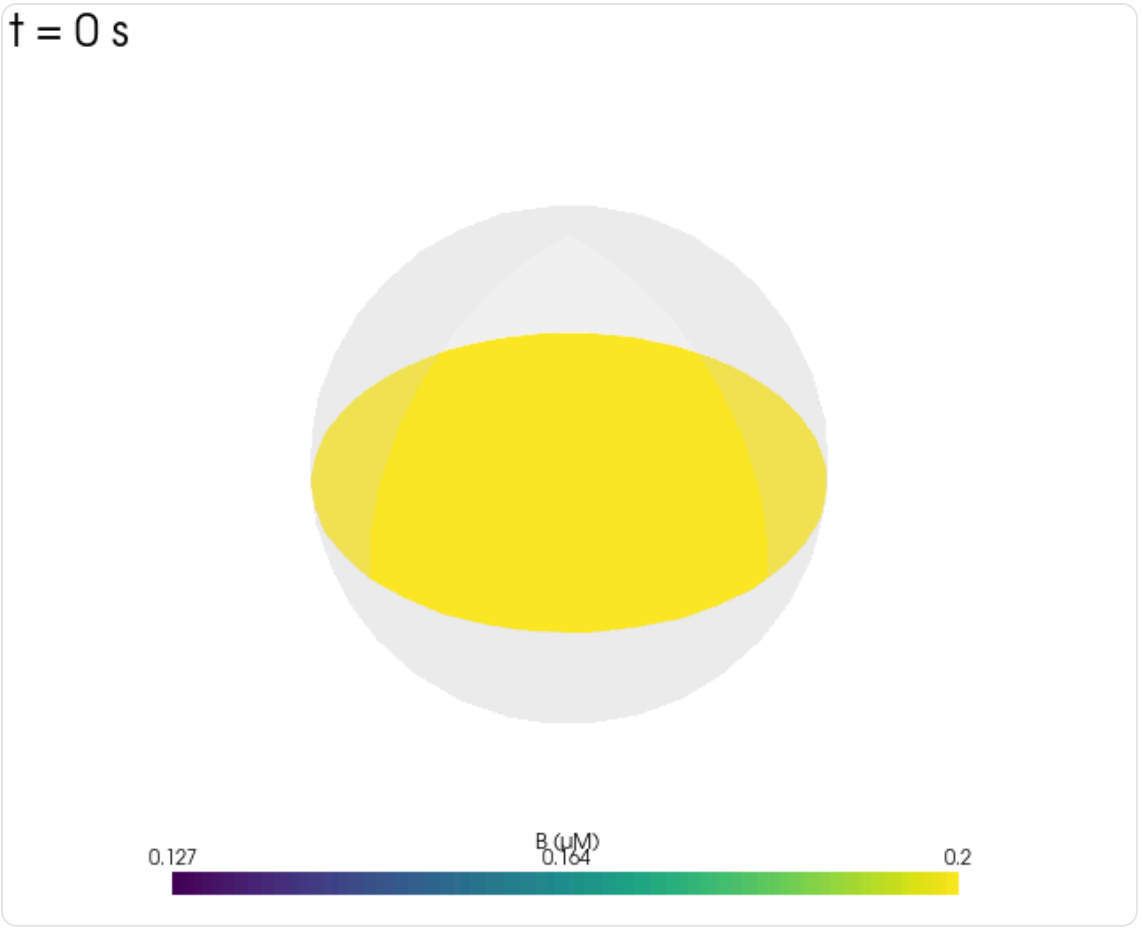
Visualizations



sphere-3d-conversion



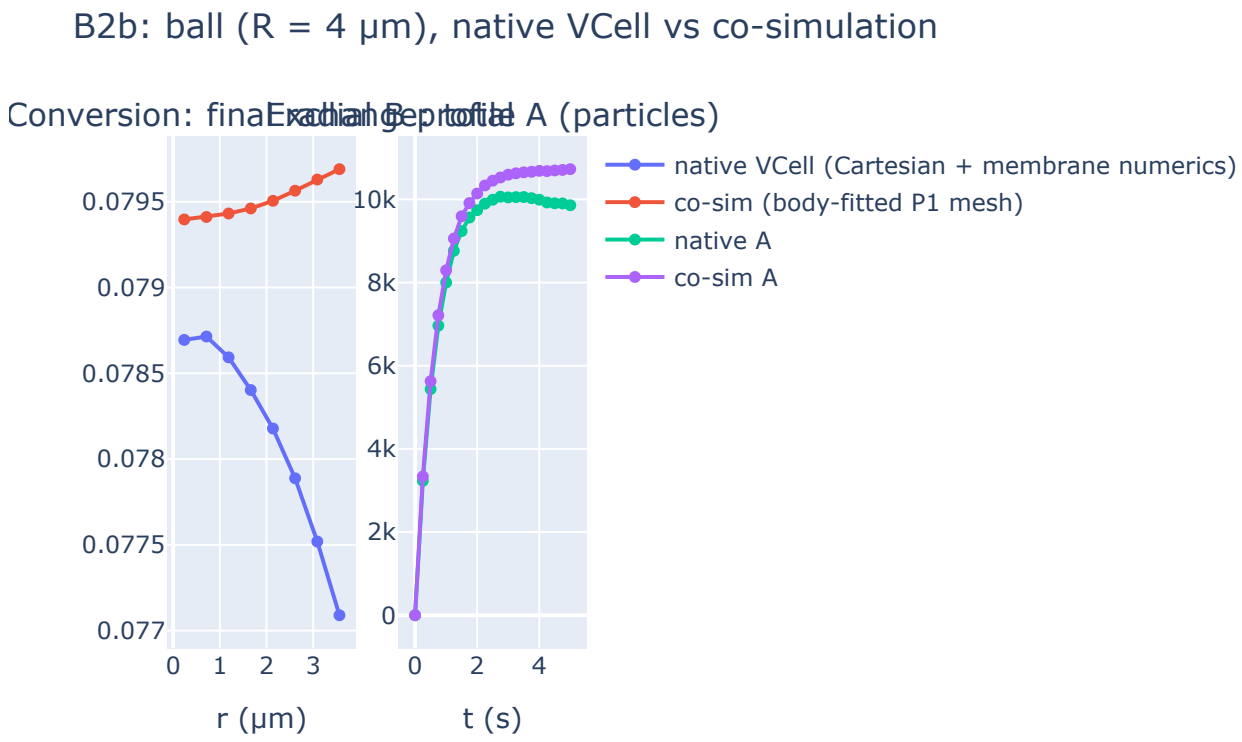
sphere-3d-exchange



sphere-3d-movie

↪ Sphere 3d interactive — too large to inline

SPHERE COSIM VS NATIVE



Finding

F-01 · confirms ● Observed

At this resolution VCell's Cartesian representation of the ball is the more accurate geometry: −0.47% volume error (b2b_native_staircase_volume_rel_err = −0.0047) against −1.4% for the co-sim body-fitted mesh at h = 0.8 μm (b2b_cosim_mesh_volume_rel_err = −0.0139).

from test: native-represents-ball-volume `range`

runs: `default`

F-02 · confirms ● Observed

For the conversion case the two discretizations agree: surviving A matches to max two-sample |z| = 1.47 (b2b_conversion_survival_max_z) and the final radial B profiles agree to 3.3% (b2b_final_radial_b_rel_diff = 0.0330, under the 5% band), with both solvers conserving A + B to under 1%.

from test: conversion-agrees `<  4`

runs: `default`

F-03 · contradicts ● Plausible

The exchange case diverges and fails its band: total A differs at max two-sample |z| = 12.1 (b2b_exchange_a_totals_max_z, band < 4.0), with native A ending 8.1% below the co-sim (b2b_exchange_native_minus_cosim_a_final_rel = −0.0808) and native signed

mass balance at −6.8% (b2b_native_exchange_mass_balance_final_signed = −0.0681) against the co-sim's +0.51%. Two native mechanisms at the curved membrane account for it: exterior binning (1.2% of A binned to nodes outside the voxelized cell, b2b_native_exchange_a_binned_outside_cell_frac = 0.0118, so their PDE source is dropped) and a field-dependent creation deficit (Smoldyn's centre-inside test skips cells that partly overlap the curved compartment). Native fixes: re-enable the same-compartment binning correction (vcell-fvsolver#25) and draw creation in every overlapping cell.

from test: exchange-agrees < 4

runs: `default`

Verdict

PASSED

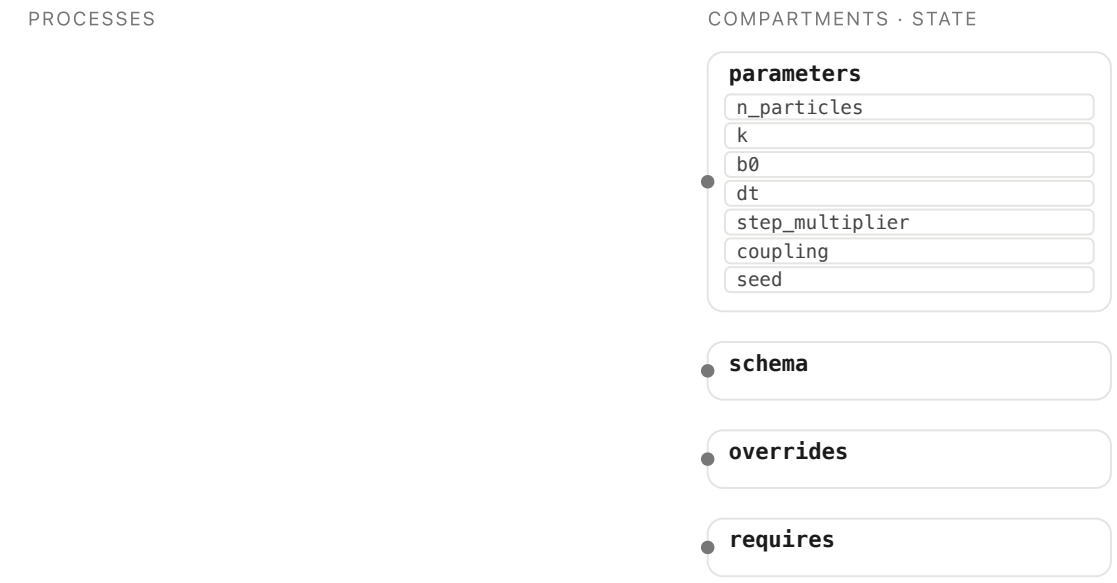
Gate cleared · Reported

Results (results/metrics.json, 2026-10-03). Re-run after Study B2d with: - the co-sim default membrane="mesh" (Smoldyn confined by the PDE mesh's boundary triangles); - the Smoldyn-fork fix for field-dependent creation in curved compartments. The native results are unchanged. **Geometry.** VCell's representation of the ball is more accurate at this resolution: −0.47% volume error, against −1.4% for the co-sim mesh at h = 0.8 µm. **Conversion: the solvers agree.** Both decay as exp(−kt) (|z| ≤ 1.5) and conserve A + B to under 1% (native −0.57%, co-sim +0.66%). Survival agrees (|z| = 1.5), and the final radial B profiles agree to 3.3%. **Exchange: the native solver loses molecules at the curved membrane.** - Signed mass balance at t = 5 s: native −6.8%; co-sim +0.51%, its first-order coupling error and within the box-domain level. - Native total A ends 8.1% below the co-sim (|z| = 12). - An earlier version of this study read the unsigned deviation as growth; that was wrong. - Two native mechanisms account for the loss. Their rough budget, about −1.6% plus −5 to −6%, matches the observed −6.8%: 1. **Exterior binning:** - 1.2% of native A is binned to grid nodes outside the voxelized cell, where B is not solved, so their PDE source k1·[A] is dropped. - B2c localizes this as a −2.3% outer-shell depletion. - In vcell-fvsolver, VCellSmoldynOutput::computeHistogram bins to the nearest node. Its same-compartment correction is commented out. 2. **Creation deficit:** - Field-dependent creation in the cell runs at 0.960 of the exact expectation. - Study B2d found the mechanism in Smoldyn's hybrid zeroreact: it draws only in grid cells whose centre is inside the compartment, so cells that partly overlap a curved compartment never create. Monte Carlo on this geometry gives 0.976 for that rule. - Our Smoldyn fork now draws in every cell and thins exactly. VCell's vendored Smoldyn 2.38 still has the centre test. - The co-sim had both effects before B2d (creation 0.974, exchange −1.2%); with the fixes, neither remains. - These are hybrid-coupling details at curved membranes, not a limitation of VCell's geometry or PDE numerics. Native-side fixes: re-enable the same-compartment binning

correction, and draw creation in every cell overlapping the compartment.

▾ Technical record — model internals, full figure set, tests, provenance & follow-ups

Model — composite internals



Static bigraph of `viva_pde_particle.composites.examples.bimolecular_hybrid` — 0 processes wired to 4 compartments. ● writes · ● reads.

`viva_pde_particle.composites.examples.bimolecular_hybrid`

Simulation run

default

`native-represents-ball-volume` authored: PASS

`conversion-agrees` authored: PASS

`conversion-field-profiles-agree` authored: PASS

`exchange-agrees` authored: FAIL

Reproduce: `vwb run study sphere-cosim-vs-native`
native-cartesian-arm: `vwb run study sphere-cosim-vs-native --variant native-cartesian-arm`
cosim-body-fitted-arm: `vwb run study sphere-cosim-vs-native --variant cosim-body-fitted-arm`

Tests

Relative volume error of VCell's Cartesian representation of the ball (cells of nodes inside). `native-represents-ball-volume · secondary · derived_scalar` Δ = 0.25 µm on R = 4 µm.	`range undefined`	pending
Max two-sample |z| of surviving A, co-sim vs native, conversion case.	`< 4`	pending

`conversion-agrees · primary · derived_scalar`

Same first-order kinetics in the same ball.

Max relative difference of the final radial B profiles (8 shells), co-sim vs native. `< 0.05` pending

`conversion-field-profiles-agree · primary · derived_scalar`

Discretization plus sampling.

Max two-sample |z| of total A, co-sim vs native, exchange case. FAILS: native loses 6.8% at the curved membrane (exterior binning plus centre-only creation in vcell-fvsolver). See the conclusion. `< 4` pending

`exchange-agrees · secondary · derived_scalar`

Kept so the open discrepancy stays visible in the report card.

FEEDBACK · THIS WHOLE STUDY 👍 Looks right ✏️ Needs revision

Add a note...

▾ 04 Fields next to a curved membrane: native VCell on pyvcell's smoothed grid vs co-sim on a body-fitted mesh

● ACCEPTED

B2b found that the native hybrid loses molecules at the curved membrane. Where does that show up in the field, and does the co-simulation have its own boundary artifact? Native fields are read on pyvcell's smoothed unstructured grid, so boundary elements sit near the true membrane rather than on the staircase.

Hypothesis & discriminating control

With A placed uniformly in the ball, A stays uniform in expectation, so E[B] is spatially uniform. Any radial structure in the ensemble-mean B therefore comes from particle-to-field transfer at the membrane. - Native: exterior-node binning drops B sources next to the membrane, so the outer shell should be depleted. - Co-sim: exterior-node counts are folded onto boundary DOFs (P^T), so its outer shell should be slightly enriched. Study B2d later revised this cause; see the conclusion.

MODEL EXPERIMENT Same ball, grid, mesh and solvers as B2b. Conversion A_p → B_f (k = 0.5 /s, 20,000 A uniform in the ball), t = 2 s, 32 seeds per solver. - E[B] = $A_0(1 - e^{(-kt)})/(V \cdot N_A)$, with V each solver's own domain volume. - The interior is r < R/2. The outer shell is r > R − 2Δ = 3.5 µm. - Native values are

volume-weighted on the smoothed cells; co-sim values are weighted by the lumped mass of the P1 DOFs.

OBSERVABLE `b2c_node_centred_smoothed_volume_rel_err, b2c_pyvcell_vtk_as_is_volume_rel_err, b2c_smoothed_surface_r_rms_err, b2c_staircase_surface_r_rms_err, b2c_native_interior_rel…`

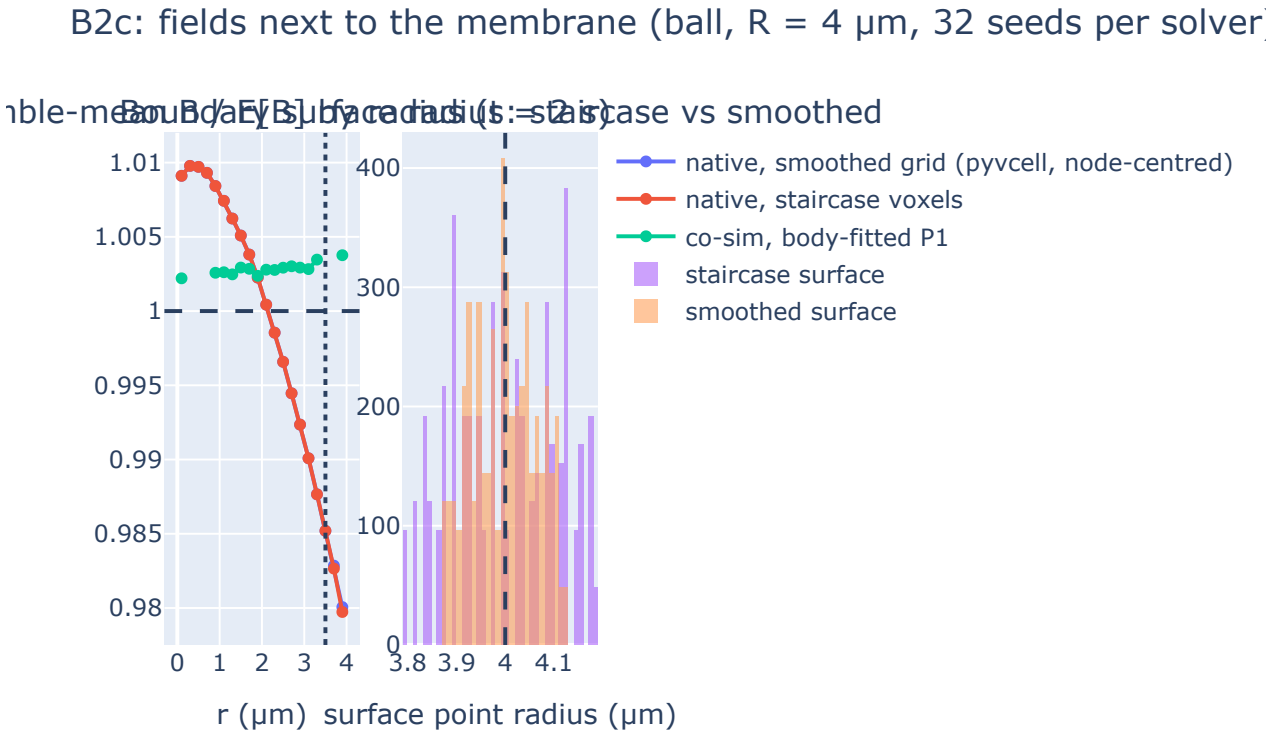
Model & key assumptions

Methods · Simulations

Model	`bimolecular_hybrid`
Sims	1 run
Tests	✓✓✓✓✓✕ 5/6 passed

Visualizations

NEAR MEMBRANE FIELDS



Finding

F-01 · confirms ● Observed

The node-centred smoothed cell domain reproduces the ball to within −0.50% (b2c_node_centred_smoothed_volume_rel_err = −0.0050), against −8.4% for the pyvcell export taken as-is, and smoothing cuts the RMS radial error of the boundary surface from 0.108 μm (staircase) to 0.065 μm.

from test: smoothed-grid-represents-ball `range`

runs: `default`

F-02 · confirms Observed

Away from the membrane both solvers match the exact uniform E[B]: the native interior is +0.49% (b2c_native_interior_rel = 0.0049) and the co-sim interior +0.25% (b2c_cosim_interior_rel = 0.0025), both well inside ±2%.

from test: native-interior-matches-expectation `range`

runs: `default`

F-03 · contradicts Plausible

The native outer shell (r > 3.5 µm) sits 2.3% below the interior (b2c_native_shell_bias = −0.0227, z = −20.1), failing the ±1% unbiasedness band, while the co-sim shell is unbiased at +0.12% (b2c_cosim_shell_bias = 0.0012, z = 1.17). This localizes B2b's exterior-binning loss to the field next to the curved membrane: molecules binned to nodes outside the voxelized cell drop their PDE source. The native fix is to re-enable the commented-out same-compartment binning correction in VCellSmoldynOutput::computeHistogram (virtualcell/vcell-fvsolver#25).

from test: native-shell-unbiased `range`

runs: `default`

Verdict

PASSED

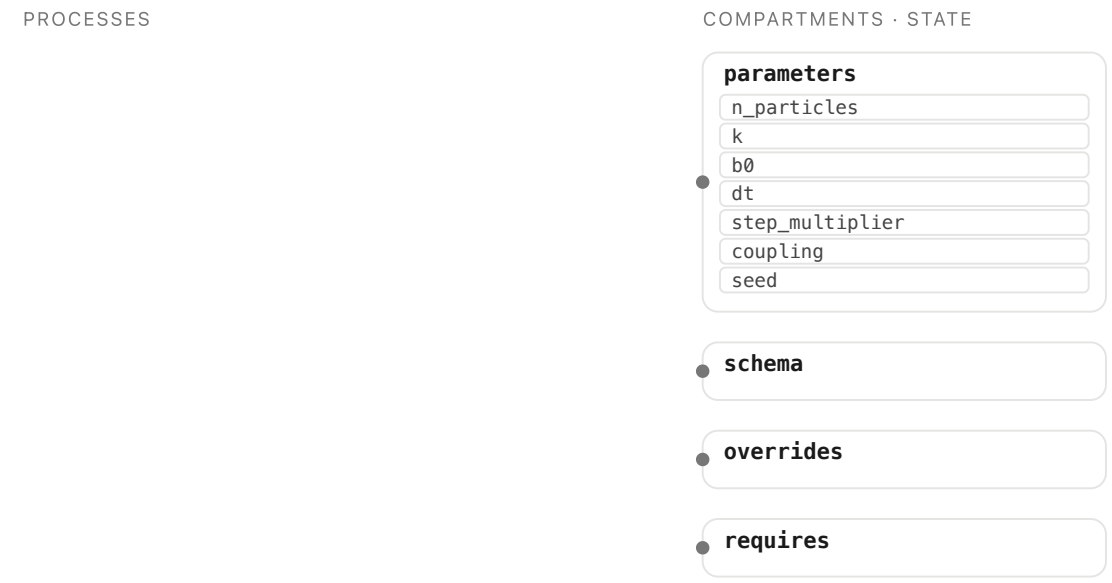
Gate cleared · Reported

Results (results/metrics.json, 2026-10-03; 32 seeds per solver; co-sim re-run with the B2d default membrane): **Geometry.** - The node-centred smoothed cell domain is within −0.50% of the ball volume. Taken as-is, the pyvcell export is −8.4% (the tiling convention above). - Smoothing cuts the RMS radial error of the boundary surface from 0.108 to 0.065 µm. **Interior.** Both solvers match the exact uniform E[B]: native +0.49%, co-sim −0.37%. **Next to the membrane.** - **Native:** - The outer shell (r > 3.5 µm) is 2.3% below the interior (z = −20). - The profile declines smoothly from about r = 2.5 µm to 0.98 at the membrane: the diffusive footprint of the B sources dropped at the boundary. - This is B2b's exterior-binning loss, now localized. - **Co-sim:** - The shell is +0.12% (z = 1.2), unbiased. - The first run, with Smoldyn confined by the exact sphere, gave +1.3% (z = 10). This study attributed that to P^T folding of exterior-node counts. Study B2d showed the cause was the domain mismatch instead: particles in the ~1.4% sliver between sphere and inscribed mesh load the boundary DOFs. - Confining Smoldyn by the mesh's own boundary (now the default) removes it. - **Smoothing** barely changes the native shell bias (−2.27% vs −2.29% on the staircase). The field is smooth on the scale of the boundary displacement (< 0.1 µm), so the smoothed grid helps geometry and visualization; the field artifact is in the coupling. Neither effect is a limitation of VCell's geometry or PDE numerics. Both come from how particles near a curved membrane are binned to the field. - Native: VCellSmoldynOutput::computeHistogram (vcell-fvsolver, bridgeVCellSmoldyn/VCellSmoldynOutput.cpp:476–548) bins each

molecule to the nearest node. Its correction for a nearest node in a different compartment (re-bin to the nearest same-compartment neighbour) is present but commented out. Re-enabling it is the likely native fix (virtualcell/vcell-fvsolver
25). - Co-sim fix (B2d, done): the mesh boundary as the particle membrane.

▾ Technical record — model internals, full figure set, tests, provenance & follow-ups

Model — composite internals



Static bigraph of `viva_pde_particle.composites.examples.bimolecular_hybrid` — 0 processes wired to 4 compartments. ● writes · ● reads.

`viva_pde_particle.composites.examples.bimolecular_hybrid`

Simulation run

default

`smoothed-grid-represents-ball` authored: PASS

`smoothing-improves-surface` authored: PASS

`native-interior-matches-expectation` authored: PASS

`cosim-interior-matches-expectation` authored: PASS

`native-shell-unbiased` authored: FAIL

`cosim-shell-unbiased` authored: PASS

Reproduce: `vwb run study near-membrane-fields`
native-smoothed-grid-arm: `vwb run study near-membrane-fields --variant native-smoothed-grid-arm`
cosim-body-fitted-arm: `vwb run study near-membrane-fields --variant cosim-body-fitted-arm`

Tests

Relative volume error of the node-centred smoothed cell domain vs the exact ball. `smoothed-grid-represents-ball · primary · derived_scalar` Whole node-centred voxels; windowed-sinc smoothing preserves volume.	`range undefined`	pending
RMS radial error of the smoothed boundary surface (µm); the staircase surface is about 0.11 µm. `smoothing-improves-surface · secondary · derived_scalar` Smoothing should cut the staircase error (Δ = 0.25 µm) substantially.	`< 0.08`	pending
Native interior (r < R/2) mean B relative to E[B]. `native-interior-matches-expectation · primary · derived_scalar` First-order conversion; the expectation is exact.	`range undefined`	pending
Co-sim interior mean B relative to E[B]. `cosim-interior-matches-expectation · primary · derived_scalar` As above.	`range undefined`	pending
Native outer-shell mean B relative to the interior. Expected to FAIL; it localizes B2b's exterior-binning loss at the membrane. `native-shell-unbiased · secondary · derived_scalar` E[B] is uniform; kept so the boundary artifact stays visible in the report card.	`range undefined`	pending
Co-sim outer-shell mean B relative to the interior. Failed (+1.3%) with the sphere membrane; passes since B2d made the mesh boundary the default membrane. `cosim-shell-unbiased · secondary · derived_scalar` As above.	`range undefined`	pending

FEEDBACK · THIS WHOLE STUDY 👍 Looks right ✏️ Needs revision

Add a note...

05 Removing the co-simulation's boundary artifact: particle transfer,

ACCEPTED

membrane geometry and curved-compartment creation

B2c found the co-simulation's ensemble-mean B 1.3% too high in the outer shell of the ball. Two causes were candidates: - the two-stage particle transfer (nearest grid node, then P^T to mesh DOFs); - the domain mismatch: Smoldyn's exact sphere holds about 1.4% more volume than the inscribed P1 mesh. Which is it, and what does removing it cost?

Hypothesis & discriminating control

B2c attributed the bias to the transfer, so the exact P1 load from molecule positions should remove it.

MODEL EXPERIMENT Same ball, background grid, mesh and model as B2b/B2c; 32 seeds per variant. Variants (particle_transfer / membrane): - grid / sphere (B2b/B2c); - positions / sphere; - grid / mesh; - positions / mesh. Cases and measures: - Conversion (uniform A, so E[B] is uniform): interior level against E[B] = $A_0(1 – e^{(–kt)})/(V_{mesh}·N_A)$, and the outer-shell bias ($r > R – 2\Delta$ vs $r < R/2$). - Exchange with field-dependent creation in the cell: final mass balance and steady ratio. - Wall time per simulated second.

OBSERVABLE `b2d_grid_mesh_shell_bias, b2d_grid_mesh_exchange_mass_balance_final_signed, b2d_grid_mesh_interior_rel, b2d_grid_sphere_shell_bias, b2d_{grid,positions}_{sphere,mesh}_wall_s_per_sim_s`

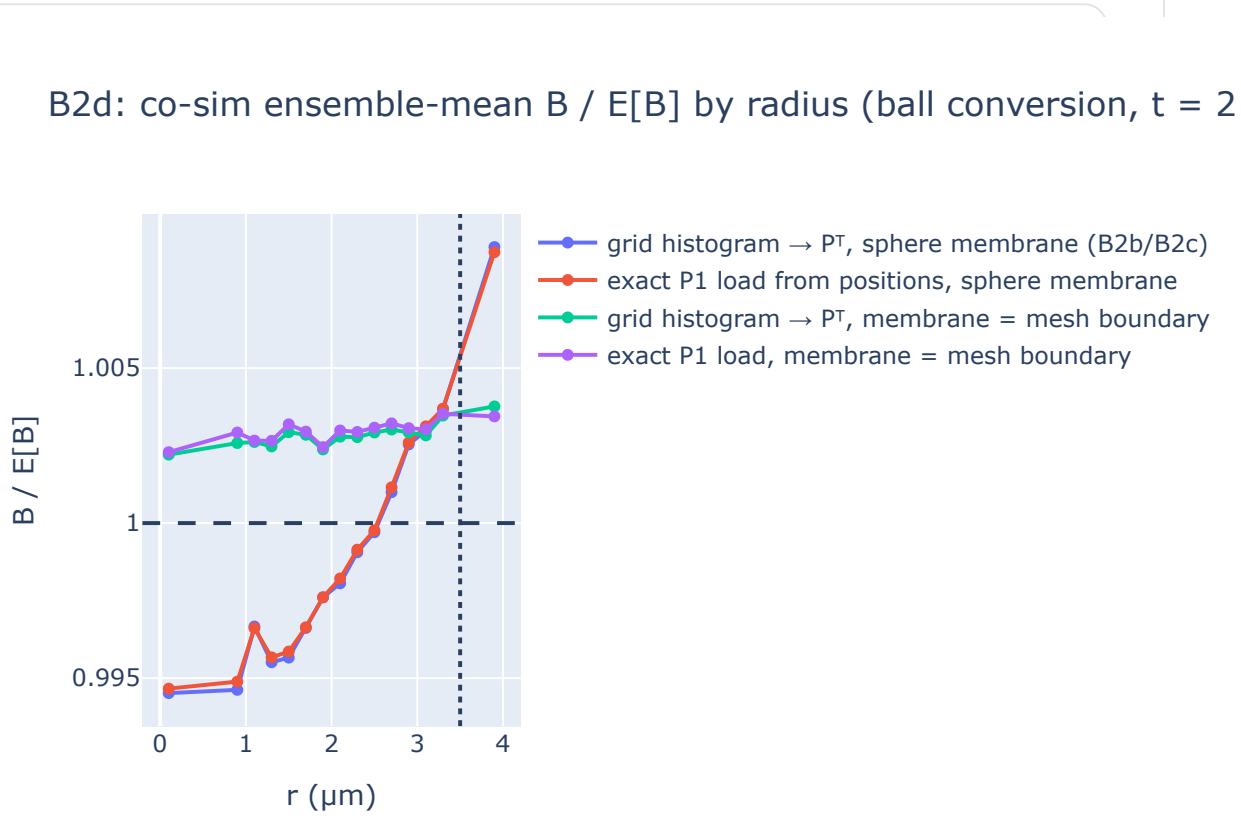
Model & key assumptions

Methods · Simulations

Model	`bimolecular_hybrid`
Sims	1 run
Tests	✓✓× 2/3 passed

Visualizations

COSIM BOUNDARY TRANSFER



Finding

F-01 · contradicts · Observed

The mesh-boundary membrane removes the outer-shell bias: with grid transfer and the mesh membrane the shell bias is +0.12% (z = 1.2), inside the ±0.5% band, down from +1.27% (z = 10.3) with the exact-sphere membrane.

from test: mesh-membrane-removes-shell-bias `range`

runs: `default`

F-02 · contradicts · Plausible

The hypothesis was wrong: the membrane, not the transfer, drives the bias. Switching grid→positions transfer under the sphere membrane leaves the shell bias at +1.24% (z = 10.0), essentially unchanged from +1.27%; only confining Smoldyn by the mesh boundary triangles removes it. The cause is the ~1.4% domain mismatch between the exact sphere and the inscribed P1 mesh.

from test: sphere-membrane-shell-bias `range`

runs: `default`

F-03 · confirms · Observed

With the mesh membrane the exchange mass balance is +0.69% (signed), inside the ±1.5% band and at the box-domain level (A3: 1.2%), so the curved geometry adds nothing once the domains match. The sphere membrane leaves it at +2.9%.

from test: mesh-membrane-exchange-balance `range`

runs: `default`

Verdict

PASSED

Gate cleared · Accepted

Results (results/metrics.json, 2026-10-03; 32 seeds per variant; Smoldyn with the creation fix): | transfer / membrane | shell bias (z) | interior vs E[B] | exchange balance | wall s per sim s | |---|---|---|---|---| | grid / sphere | +1.27% (10.3) | −0.37% | +2.9% | 0.71 | | positions / sphere | +1.24% (10.0) | −0.36% | +2.9% | 2.98 | | grid / mesh | +0.12% (1.2) | +0.25% | +0.69% | 1.07 | | positions / mesh | +0.07% (0.7) | +0.27% | +1.0% | 3.56 | **The hypothesis was wrong: the membrane matters, not the transfer.** - With the sphere membrane, the exact P1 load leaves the shell bias unchanged. - The cause is the domain mismatch. Particles in the ~1.4% sliver between the sphere and the inscribed mesh are projected onto the boundary DOFs. - Confining Smoldyn by the mesh's own boundary triangles removes the bias, with either transfer. VCell does the same with its geometry's surface triangulation; see below. - With the mesh membrane, exchange mass balance is at the box-domain level (A3: 1.2%), so the curved geometry adds nothing. - With the sphere membrane, the PDE removes B over the smaller mesh while creation covers the full sphere, so the balance is +2.9%. **Credit: VCell already does this.** Its Smoldyn input confines particles by `panel tri` surfaces taken from its own geometry (B2b ball: 9,516 triangles enclosing 261.8 µm³), so the co-sim's mesh membrane follows VCell's practice. - **Co-sim:** the triangles are the PDE mesh boundary itself, so the domains are identical. - **Native:** the triangulation is a smoothed surface, while the PDE domain is the voxel staircase (266.7 µm³). The particle compartment is therefore 1.8% smaller than the PDE's cell. Times the centre-only creation factor (0.976), that predicts a native creation rate of 0.958, matching the measured 0.960. Particles near the smooth membrane can also be nearest to exterior voxel nodes (exterior binning). **Creation fix (Smoldyn fork).** Before it, the sphere variants showed −1.2 to −1.4% exchange balance. That was two errors cancelling: a creation deficit of about −2.4% and the sphere's extra +1.4% of volume. The mesh membrane exposed the deficit (−3.2%); the fix removes it. **Cost.** - The wall times in the table are for the conversion case. - Positions transfer costs about 4× (locating 20k particles every step), and buys nothing here once the membrane matches. - Follow-up: exchange with the mesh membrane was much slower, about 23× the sphere. - Smoldyn sized its virtual boxes from the initial molecule count (zero A), giving one box, so every molecule tested all 806 triangles every step (N×M). - The Smoldyn config now always sets boxsize, which gives 10×. - It also writes VCell's highResVolumeSamples compartment map (fork e86d552 enables its parsing under OPTION_VCELL), a further 1.8×, with identical results. - Mesh-membrane exchange now runs at 0.25 s per simulated s, against 0.21 for the sphere; see DESIGN.md §11. **Default changed:** build_mesh_hybrid_document now uses

membrane="mesh" with grid transfer. B2, B2b and B2c are re-run with it.

▾ Technical record — model internals, full figure set, tests, provenance & follow-ups

Model — composite internals



Static bigraph of `viva_pde_particle.composites.examples.bimolecular_hybrid` — 0 processes wired to 4 compartments. ● writes · ● reads.

`viva_pde_particle.composites.examples.bimolecular_hybrid`

Simulation run

default

`mesh-membrane-removes-shell-bias` authored: PASS

`mesh-membrane-exchange-balance` authored: PASS

`sphere-membrane-shell-bias` authored: FAIL

Reproduce: `vwb run study cosim-boundary-transfer`
grid-sphere: `vwb run study cosim-boundary-transfer --variant grid-sphere`
positions-sphere: `vwb run study cosim-boundary-transfer --variant positions-sphere`
grid-mesh: `vwb run study cosim-boundary-transfer --variant grid-mesh`
positions-mesh: `vwb run study cosim-boundary-transfer --variant positions-mesh`

Tests

Outer-shell bias of the default co-sim (grid transfer, mesh membrane). `range undefined` pending

`mesh-membrane-removes-shell-bias · primary · derived_scalar`

E[B] is uniform; 32 seeds give an SE of about 0.1%.

Exchange mass balance of the default co-sim at t = 5 s, signed.
`mesh-membrane-exchange-balance · primary · derived_scalar`
First-order coupling error only. The box-domain baseline (A3, two-way-exchange) is 1.2%.

range undefined pending

Outer-shell bias with the exact-sphere membrane. Expected to FAIL; it documents the domain mismatch.
`sphere-membrane-shell-bias · secondary · derived_scalar`
Kept so the artifact stays visible in the report card.

range undefined pending

FEEDBACK · THIS WHOLE STUDY 👍 Looks right ✏️ Needs revision

Add a note...

06 The co-simulation on VCell's own geometry: staircase PDE, smooth membrane, accessible-volume correction

ACCEPTED

Native VCell solves the PDE on the voxel staircase but confines particles by a smooth membrane. For the B2b ball the two domains differ by ~1.8% in volume. That leaves a −1.4% outer-shell depletion and an exchange drift, even with the binning correction of vcell-fvsolver#25 (Study B2c). With the co-simulation running on exactly VCell's geometry, can an accessible-volume correction make the coupling both unbiased and conservative?

Hypothesis & discriminating control

Three variants of the particle → field transfer: - **`none`** (full voxel volumes): reproduces native VCell with the
25 correction. The shell is depleted by ~1.4%, and exchange drifts by the smooth-vs-staircase volume mismatch. - **`adapters`** (accessible volumes in the adapter only): removes the shell bias, but the PDE still integrates over full voxels, so particle sources are overcounted (~+1.4% in conversion). - **`adapters+volumes`** (accessible volumes in the adapter and as the FV element volumes): unbiased and conservative.

MODEL EXPERIMENT The B2b ball as a vcell-fenics GeometryDescription, realized by VCell itself (VCellFVRealization, via libvcell): - **PDE:** FV on VCell's staircase

`cell` (266.734 µm³). - **Smoldyn:** VCell's smooth membrane (9,516 triangles enclosing 261.818 µm³) and VCell's compartment points. - **Adapters:** counts at exterior nodes are folded into the domain (compartment-aware binning), and fields are extended into the exterior band for 0th-order creation. 32 seeds per variant. - Conversion (k = 0.5, 20,000 A uniform in the smooth cell): interior B against E[B] = $A_0(1 - e^{(-kt)})/(V_PDE \cdot N_A)$, the outer-shell bias (r > 3.5 vs r < 2) and A + B balance. - Exchange (k1 = 1, k2 = 0.5, B_0 = 0.2 µM): A + B balance and the steady ratio. All PDE masses and averages use the PDE's own element volumes.

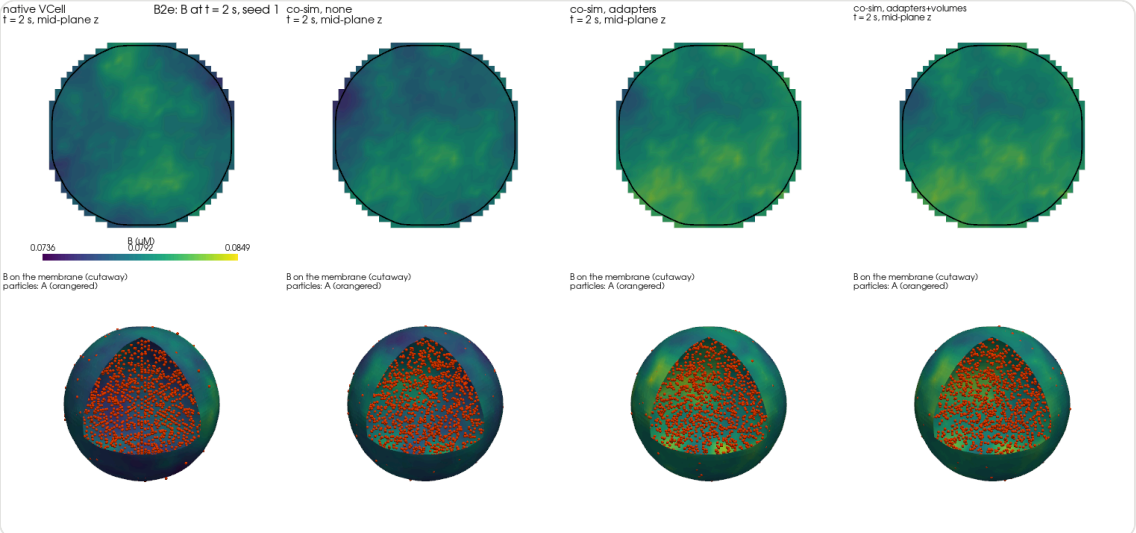
OBSERVABLE `b2e_adapters_volumes_shell_bias, b2e_adapters_volumes_interior_rel, b2e_adapters_volumes_exchange_balance, b2e_adapters_volumes_exchange_steady_ratio_rel_err, b2e_none_shell_bias…`

Model & key assumptions

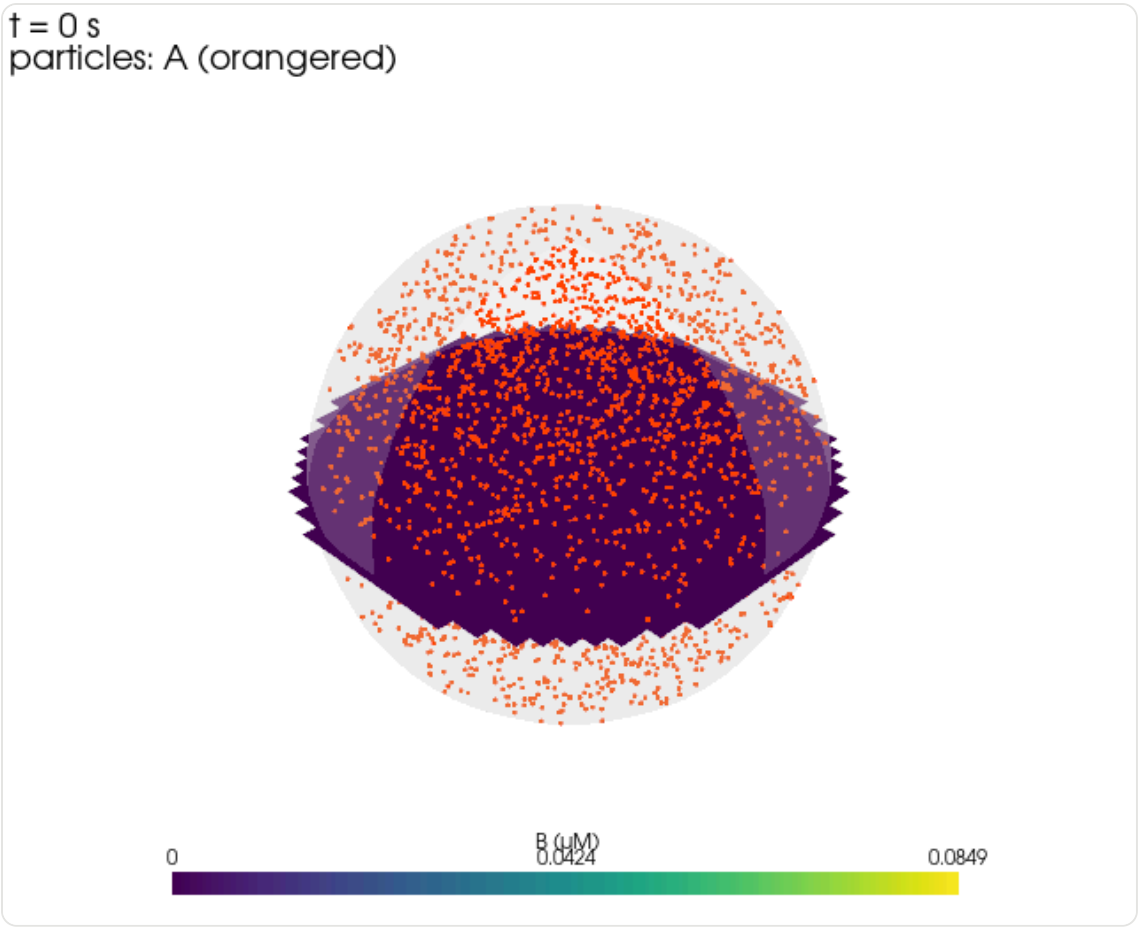
Methods · Simulations

Model	`bimolecular_hybrid`
Sims	1 run
Tests	✓✓✓✕ 3/4 passed

Visualizations



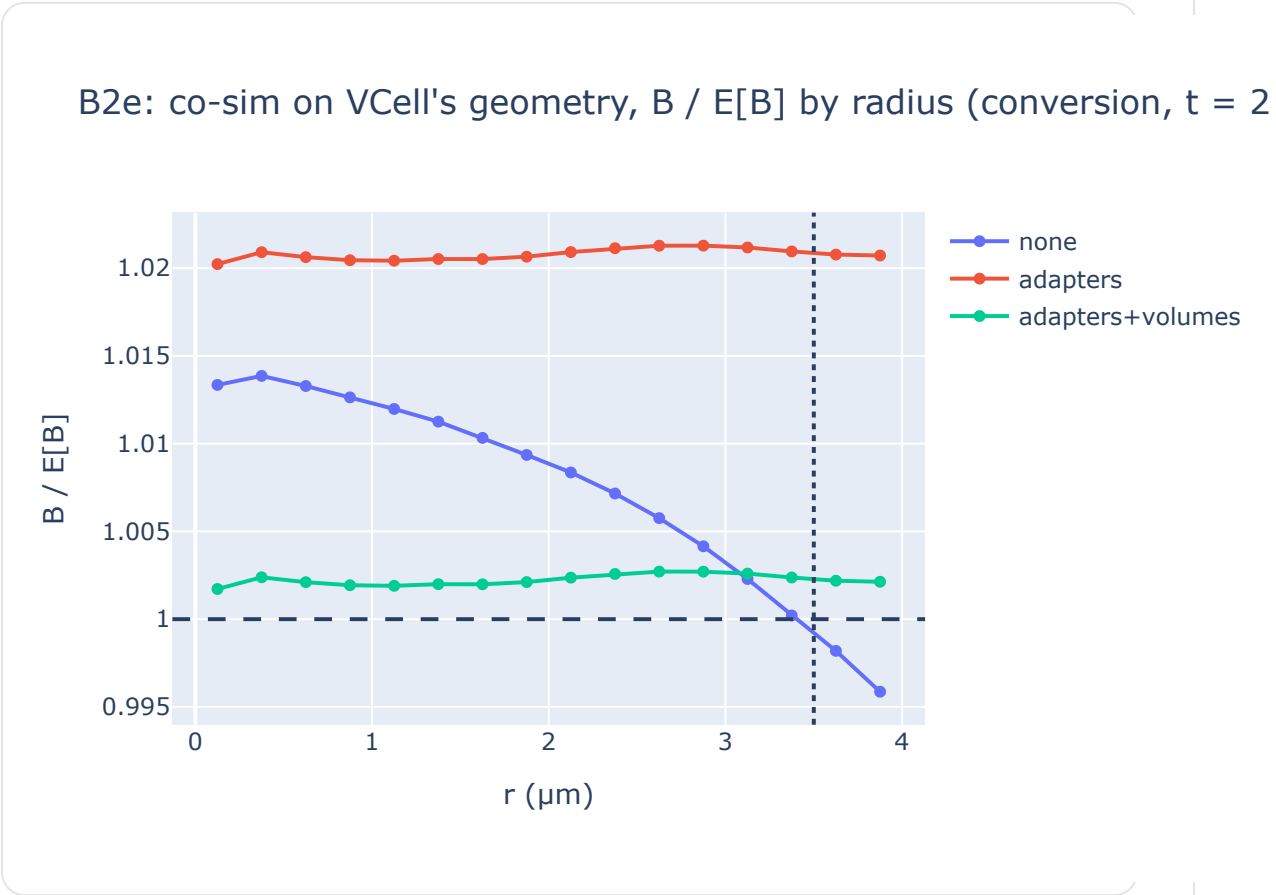
vcell-geometry-3d



vcell-geometry-3d-movie

↪ Vcell geometry 3d interactive — too large to inline

VCELL GEOMETRY COSIM



Finding

F-01 · contradicts · Plausible

The adapters+volumes correction makes the coupling unbiased on VCell's own geometry: the outer-shell bias is +0.017% (b2e_adapters_volumes_shell_bias = 0.000167, within ±0.5%) and the interior is +0.20% of E[B] (b2e_adapters_volumes_interior_rel = 0.00203, within ±1%). Using accessible volumes both in the adapter and as the FV element volumes removes the smooth-vs-staircase volume mismatch.

from test: corrected-shell-unbiased `range`

runs: `default`

F-02 · confirms · Observed

The corrected coupling stays conservative in exchange: A + B balance at t = 5 s is +0.96% (b2e_adapters_volumes_exchange_balance = 0.00959, within ±1.5% and below the 1.2% box-domain A3 baseline), and the steady ratio error is 0.13% (b2e_adapters_volumes_exchange_steady_ratio_rel_err = 0.00128) — the best of the three corrections.

from test: corrected-exchange-conserves `range`

runs: `default`

F-03 · contradicts · Observed

The intentional regression check FAILS as designed: with correction="none" (full voxel volumes) the outer-shell bias is −1.36% (b2e_none_shell_bias = −0.0136, z = −12.5), outside the ±0.5% band. This reproduces native VCell with the vcell-fvsolver#25 binning correction (native −1.41%, forward model −1.33%) and keeps the smooth-vs-staircase volume mismatch (staircase 266.73 µm³ vs smooth 261.82 µm³, ~1.8%) visible in the report card.

from test: vcell-like-shell-depletion `range`

runs: `default`

Verdict

PASSED

Gate cleared · Accepted

Results (results/metrics.json, 2026-10-04; 32 seeds per variant): | variant | shell vs interior (z) | interior vs E[B] | conversion balance | exchange balance | steady ratio err | |---|---|---|---|---|---| | none | −1.36% (−12.5) | +1.07% | +0.12% | −2.7% | 1.2% | | adapters | +0.02% (0.2) | +2.06% | +1.30% | +0.19% | 1.8% | | adapters+volumes | +0.02% (0.15) | +0.20% | +0.12% | +0.96% | 0.13% | - **`none` reproduces native VCell with the vcell-fvsolver 25 binning correction. The shell is** −1.36% (native −1.41%; the forward model −1.33%) and the interior +1.07% (native +1.14%). Exchange drifts −2.7%: creation runs over the smooth volume, while consumption runs over the staircase volume. - **`adapters`** removes the shell bias but overcounts particle sources (+1.30% in conversion). In exchange this overcount happens to cancel the creation mismatch,

which is not a principled result. - **`adapters+volumes` is unbiased and conservative.** The shell bias is +0.02% (z = 0.15); the interior is within 0.2% of E[B]; conversion is at the lag level (+0.12%) and exchange at the first-order coupling level (+0.96%, against 1.2% for the box-domain A3). It also has the best steady ratio. - Native vcell-fvsolver keeps full voxel volumes. The co-simulation offers both: `none` to compare with VCell, and `adapters+volumes` (the default) for accuracy.

▾ **Technical record — model internals, full figure set, tests, provenance & follow-ups**

Model — composite internals



Static bigraph of `viva_pde_particle.composites.examples.bimolecular_hybrid` — 0 processes wired to 4 compartments. ● writes · ● reads.

`viva_pde_particle.composites.examples.bimolecular_hybrid`

Simulation run

default

`corrected-shell-unbiased` authored: PASS

`corrected-interior-matches` authored: PASS

`corrected-exchange-conserves` authored: PASS

`vcell-like-shell-depletion` authored: FAIL

Reproduce: `vwb run study vcell-geometry-cosim`
correction-none: `vwb run study vcell-geometry-cosim --variant correction-none`
correction-adapters: `vwb run study vcell-geometry-cosim --variant correction-adapters`
correction-adapters-volumes: `vwb run study vcell-geometry-cosim --variant correction-adapters-volumes`

Tests

Outer-shell mean B relative to the interior with correction="adapters+volumes". `corrected-shell-unbiased · primary · derived_scalar` E[B] is spatially uniform; 32 seeds give an SE of about 0.1%.	`range undefined`	pending
Interior mean B relative to E[B] with correction="adapters+volumes". `corrected-interior-matches · primary · derived_scalar` First-order conversion; the expectation is exact on the PDE's own volume.	`range undefined`	pending
A + B balance at t = 5 s (exchange) with correction="adapters+volumes". `corrected-exchange-conserves · primary · derived_scalar` First-order coupling error only; the box-domain baseline (A3) is 1.2%.	`range undefined`	pending
Outer-shell bias with correction="none" (VCell's full voxel volumes). Expected to FAIL; it reproduces native VCell with the #25 binning correction (B2c: −1.41%). `vcell-like-shell-depletion · secondary · derived_scalar` Kept so the volume mismatch stays visible in the report card.	`range undefined`	pending

FEEDBACK · THIS WHOLE STUDY 👍 Looks right ✏️ Needs revision

Add a note...

▾ 07

One non-convex GeometryDescription, three solver paths: native VCell, co-sim on VCell's geometry, co-sim on a Netgen mesh

● ACCEPTED

Phase 7e end to end on a non-sphere: does one geometry description drive the native VCell solver and both co-simulation paths, and do the accessible-volume correction and the body-fitted mesh stay unbiased and conservative on a thin-necked, non-convex shape?

Hypothesis & discriminating control

The ball's ~1.8% smooth-vs-staircase mismatch grows with surface-to-volume ratio and thin features, so the uncorrected paths (native VCell; the co-sim with

none) should degrade much more on a dumbbell. `adapters+volumes` and the Netgen mesh should stay at the coupling-lag level.

MODEL EXPERIMENT A vcell-fenics GeometryDescription: two balls (R = 1.5 µm, centres 2.8 µm apart) joined by a neck (radius 0.6 µm), in an 8 × 6 × 6 µm box, Δ = 0.25 µm (33 × 25 × 25 grid). The four paths: - **native:** the description converted to VCML; vcell-fvsolver with embedded Smoldyn; 8 seeds. - **vcell_none, vcell_corrected:** the co-simulation on VCell's own realization, with correction `none` or `adapters+volumes`; 16 seeds each. - **mesh:** the co-simulation on a Netgen tet mesh (h = 0.3); 16 seeds. Cases and measures: - conversion (k = 0.5, 20,000 A uniform; 2 s): A + B balance; near-membrane bias (mean B within 0.5 µm of the membrane against deeper B; E[B] is uniform); lobe asymmetry; - exchange (k1 = 1, k2 = 0.5, B_0 = 0.2 µM; 5 s): A + B balance and the steady ratio.

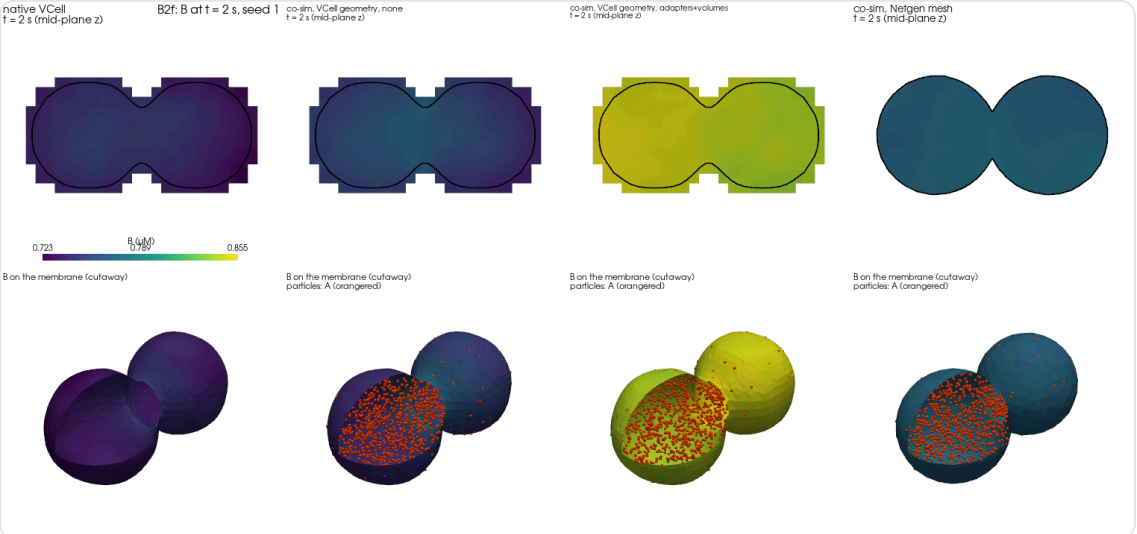
OBSERVABLE `b2f_vcell_corrected_exchange_balance, b2f_vcell_corrected_band_bias, b2f_mesh_exchange_balance, b2f_native_exchange_balance, b2f_volume_vcell_smooth_rel`

Model & key assumptions

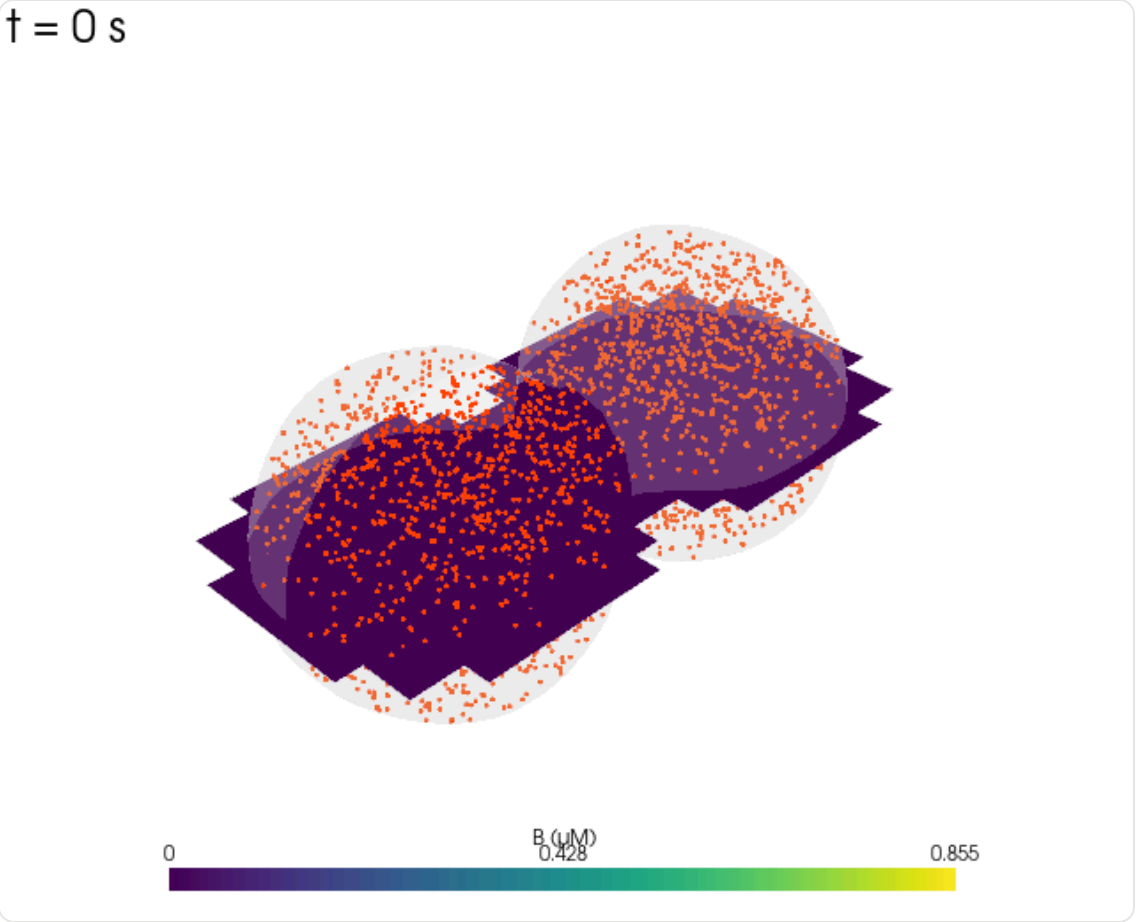
Methods · Simulations

Model	`bimolecular_hybrid`
Sims	1 run
Tests	✓✓✓✕ 3/4 passed

Visualizations



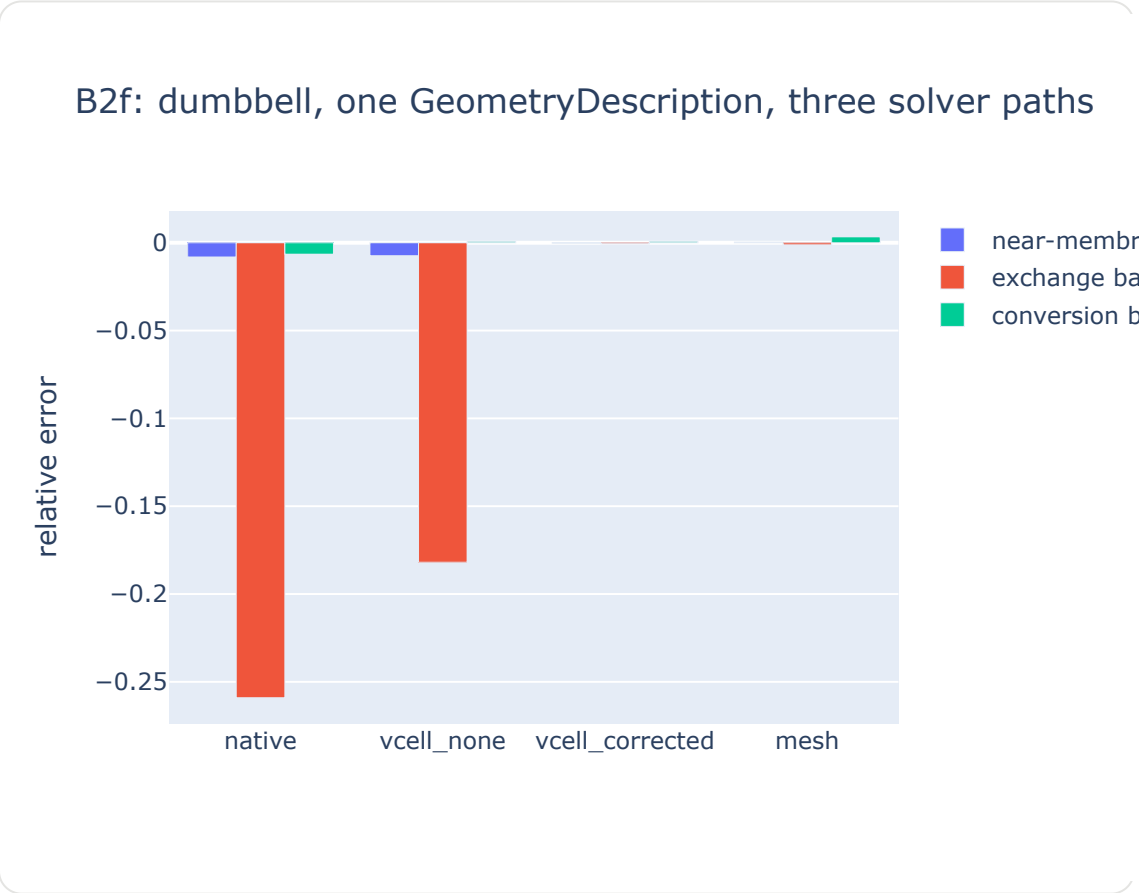
dumbbell-3d



dumbbell-3d-movie

DUMBBELL 3D INTERACTIVE

DUMBBELL THREE WAYS



Finding

F-01 · contradicts · Observed

The accessible-volume correction keeps the co-sim conservative and unbiased on exactly VCell's thin-necked geometry: exchange A+B balance is -0.07% (±1.5% band) and the near-membrane bias is -0.02% (z = -1.2, ±0.5% band), with the steady ratio off by 0.39%.

from test: corrected-exchange-conserves `range`

runs: `default`

F-02 · contradicts · Observed

The body-fitted Netgen mesh path is also conservative: exchange balance is -0.12% (±1.5% band) with a +0.04% near-membrane bias, against a -3.0% mesh cell-volume error.

from test: mesh-exchange-conserves `range`

runs: `default`

F-03 · contradicts · Plausible

On a thin-necked shape the uncorrected smooth-vs-staircase mismatch is severe and the native path fails the band: native VCell loses -25.9% of A+B in exchange and misses the steady ratio by 8.9%, driven by the -12.4% smooth-membrane volume error that shrinks the neck and small lobes. The co-sim with VCell's own volumes (`none`) isolates the volume effect at -18.2%; the rest is consistent with the binning (vcell-fvsolver#25) and creation (#26) defects.

from test: native-exchange-conserves `range`

runs: `default`

Verdict

PASSED

Gate cleared · Accepted

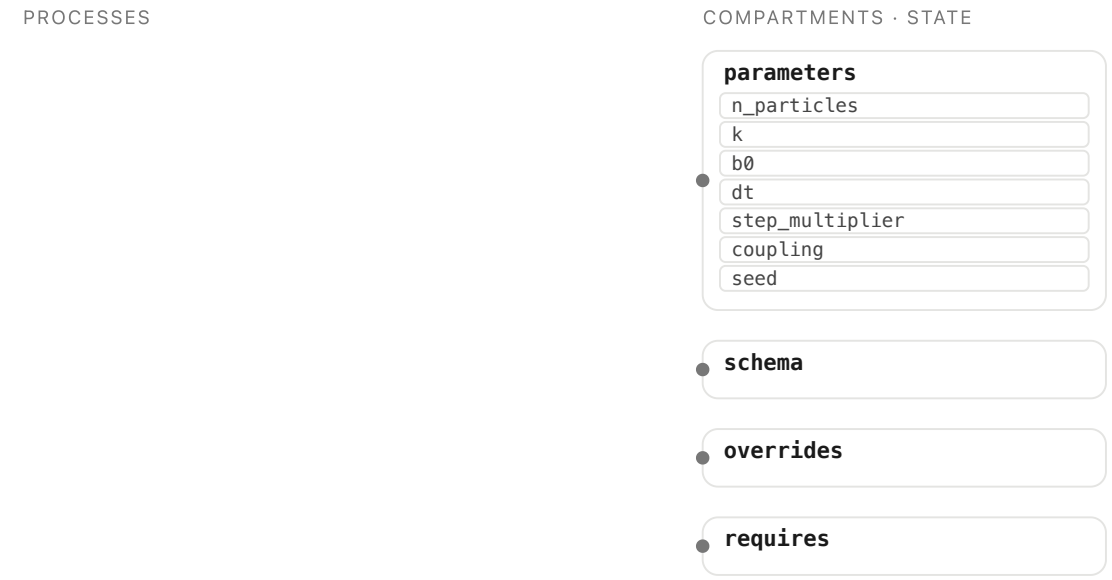
Results (results/metrics.json, 2026-10-04). The analytic cell volume is 28.11 µm³ (Monte Carlo). **Geometry:** - VCell's staircase is –1.6%. - VCell's smooth membrane is **–12.4%**: smoothing shrinks the thin neck and the small lobes. - The Netgen mesh (h = 0.3) is –3.0%. | path | near-membrane bias (z) | lobe asym (z) | conversion | exchange | steady ratio err | |---|---|---|---|---|---| | native VCell | –0.81% (–118) | +0.07% (0.2) | –0.65% | **–25.9%** | 8.9% | | co-sim, VCell geometry, none | –0.74% (–43) | +0.15% (0.4) | +0.06% | **–18.2%** | 7.8% | | co-sim, VCell geometry, adapters+volumes | –0.02% (–1.2) | +0.33% (0.9) | +0.06% | –0.07% | 0.39% | | co-sim, Netgen mesh | +0.04% (2.8) | +0.60% (1.9) | +0.34% | –0.12% | 0.61% | - **On a thin-necked shape the uncorrected smooth-vs-staircase mismatch is severe.** Native VCell loses a quarter of A + B in exchange and misses the steady ratio by 9%. - The co-sim with VCell's own volumes (none) isolates the volume effect: –18.2%. - The rest of the native loss is consistent with the binning (vcell-fvsolver

25) and creation (

26) defects measured on the ball. - **The accessible-volume correction removes it entirely.** Exchange is –0.07%, the near-membrane bias is gone and the steady ratio is accurate, on exactly VCell's geometry. - **The body-fitted Netgen path** is also conservative and unbiased. Its slight lobe asymmetry (0.6%, z = 1.9) is borderline at 16 seeds. - One description drove all four paths; no solver-specific geometry code was needed.

▾ Technical record — model internals, full figure set, tests, provenance & follow-ups

Model — composite internals



Static bigraph of `viva_pde_particle.composites.examples.bimolecular_hybrid` — 0 processes wired to 4 compartments. ● writes · ● reads.

`viva_pde_particle.composites.examples.bimolecular_hybrid`

Simulation run

default

`corrected-exchange-conserves` authored: PASS

`corrected-near-membrane-unbiased` authored: PASS

`mesh-exchange-conserves` authored: PASS

`native-exchange-conserves` authored: FAIL

Reproduce: `vwb run study dumbbell-three-ways`
native: `vwb run study dumbbell-three-ways --variant native`
vcell_none: `vwb run study dumbbell-three-ways --variant vcell_none`
vcell_corrected: `vwb run study dumbbell-three-ways --variant vcell_corrected`
mesh: `vwb run study dumbbell-three-ways --variant mesh`

Tests

A + B balance at 5 s, co-sim on VCell's geometry with adapters+volumes. `range undefined` pending

corrected-exchange-conserves · primary · derived_scalar

First-order coupling error only.

Near-membrane B bias, co-sim on VCell's geometry with adapters+volumes. corrected-near-membrane-unbiased · primary · derived_scalar E[B] is spatially uniform.	range undefined	pending
A + B balance at 5 s, co-sim on the Netgen mesh. mesh-exchange-conserves · primary · derived_scalar The particle domain is the mesh (B2d).	range undefined	pending
A + B balance at 5 s, native VCell. Expected to FAIL; it documents the smooth-vs-staircase mismatch plus the vcell-fvsolver#25/#26 defects on a thin-necked shape. native-exchange-conserves · secondary · derived_scalar Kept so the native defect stays visible in the report card.	range undefined	pending

FEEDBACK · THIS WHOLE STUDY 👍 Looks right ✏️ Needs revision

Add a note…

▾ 08 Operator-splitting schemes for PDE/particle coupling: Jacobi (fvsolver), Gauss–Seidel, Strang

● ACCEPTED

The embedded vcell-fvsolver hard-codes one lagged (Jacobi-type) splitting. With the coupling expressed as process-bigraph processes, can other splittings be used, and how much accuracy do they buy at a given coupling interval?

Hypothesis & discriminating control

Jacobi and both Gauss–Seidel orderings are first order in the coupling interval τ. Symmetric Strang splitting (half the PDE substeps, the particle step, the other half) is second order in the mean dynamics and much more accurate at large τ.

MODEL EXPERIMENT A5's A_p + B_f → C_f (k = 2/(μM·s), 20,000 A, 11×11×3 slab); Δt = 0.005; τ = k·Δt with k ∈ {2, 4, 8, 16, 32}; 64 seeds per point. Error: max over time of |mean survival − continuum|. Orders fitted for k ≥ 4.

OBSERVABLE `b3_jacobi_order, b3_gs_particles_first_order, b3_gs_pde_first_order, b3_strang_order, b3_strang_over_jacobi_err_k32…`

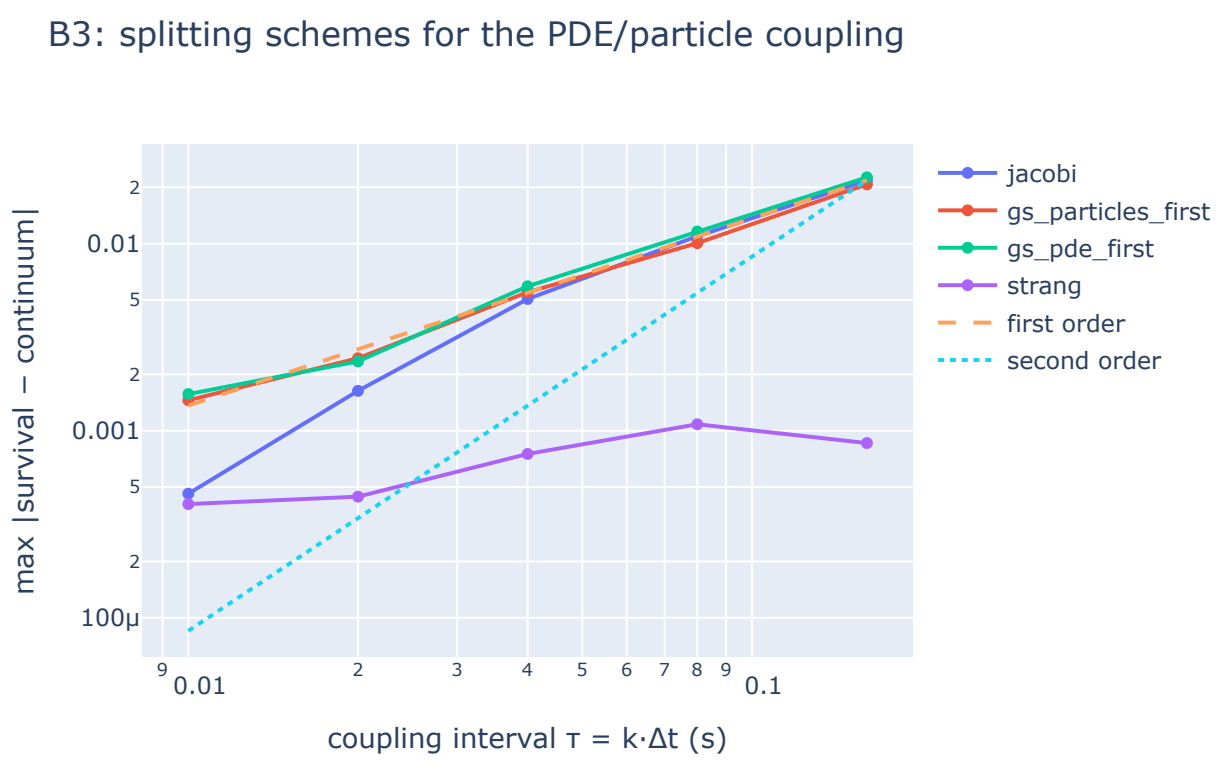
Model & key assumptions

Methods · Simulations

Model	`bimolecular_hybrid`
Sims	1 run
Tests	✓✓✓ 3/3 passed

Visualizations

SPLITTING SCHEMES



Finding

F-01 · confirms ● Observed

All three lagged splittings are first order in the coupling interval τ: the fitted order is 1.23 for Jacobi (fvsolver), 1.01 for Gauss–Seidel (particles-first) and 1.08 for Gauss–Seidel (pde-first). At τ = 0.16 s their errors cluster at 2.1–2.3% (Jacobi 0.0218, GS-particles 0.0208, GS-pde 0.0226).

from test: jacobi-first-order `range`

runs: `default`

F-02 · confirms ● Observed

Symmetric Strang splitting is dramatically more accurate at large coupling interval: at k = 32 (τ = 0.16 s) the Strang error is 0.000859, 3.94% of the Jacobi error

(b3_strang_over_jacobi_err_k32 = 0.0394, i.e. a 25× reduction), well under the 0.25 acceptance band. Expressing the coupling as process-bigraph substeps makes this re-ordering a coupler option rather than a solver modification.

from test: strang-much-more-accurate < `0.25`

runs: `default`

F-03 · partial ● Observed

Strang's error stays within ~2× of the sampling floor across the whole sweep (0.000405 at k=2 to 0.000859 at k=32), so its fitted order (0.34) is floor-dominated and not resolvable — more seeds or larger τ would be needed to measure the expected second order.

from test: strang-much-more-accurate < `0.25`

runs: `default`

Verdict

● PASSED **Gate cleared** · Accepted

Results (results/metrics.json, 2026-10-03): - **First-order schemes:** Jacobi (fvsolver) has order 1.23, Gauss–Seidel orders are 1.01 and 1.08. At τ = 0.16 s their errors are about 0.021–0.023. - **Strang:** the error stays within about 2× of the sampling floor ($4–11 \cdot 10^{-4}$) for every τ up to 0.16 s, 25× smaller than Jacobi at τ = 0.16 s. It is too close to the floor to fit an order; more seeds or larger τ would resolve it. - **Consequence:** for this model, Strang splitting lets the particle side step about 32× less often than the PDE side at roughly the accuracy fvsolver's scheme reaches with k ≈ 1–2. That's a large potential speed-up for particle-dominated runs (A6). vcell-fvsolver's hybrid loop (SimTool.cpp) fixes one splitting order, so adopting Strang there would mean modifying the solver; here it is a coupler option.

▾ Technical record — model internals, full figure set, tests, provenance & follow-ups

Model — composite internals



Static bigraph of `viva_pde_particle.composites.examples.bimolecular_hybrid` — 0 processes wired to 4 compartments. ● writes · ● reads.

`viva_pde_particle.composites.examples.bimolecular_hybrid`

Simulation run

default

`jacobi-first-order` authored: PASS

`gauss-seidel-first-order` authored: PASS

`strang-much-more-accurate` authored: PASS

Reproduce: `vwb run study splitting-schemes`
splitting-scheme-sweep: `vwb run study splitting-schemes --variant splitting-scheme-sweep`
coupling-interval-sweep: `vwb run study splitting-schemes --variant coupling-interval-sweep`

Tests

Fitted order of the Jacobi (fvsolver) scheme. — `range undefined` pending
`jacobi-first-order · primary · derived_scalar`
Explicit lagged splitting is first order.

Fitted order of Gauss–Seidel (particles first). — `range undefined` pending
`gauss-seidel-first-order · secondary · derived_scalar`
Sequential (Lie) splitting is first order.

Strang error / Jacobi error at the largest coupling interval (k = 32, τ = 0.16 s). — `< 0.25` pending
`strang-much-more-accurate · primary · derived_scalar`
Symmetric splitting cancels the leading splitting error.

FEEDBACK · THIS WHOLE STUDY 👍 Looks right ✏️ Needs revision

Add a note…

09 The hybrid model described in SBML Spatial, run on both solvers

ACCEPTED

Can a PDE/particle hybrid be described in SBML Spatial, with each species' representation (continuous or particle) as a VCell annotation, and run unchanged from that SBML on both the native VCell solver and the co-simulation?

Hypothesis & discriminating control

The SBML carries everything the model needs (geometry, species, diffusion, initial conditions, mass-action reactions, representations), so a run from the SBML reproduces the run from the original model exactly, seed for seed, on each solver.

MODEL EXPERIMENT The A3 two-way exchange (11×11×3 slab, 10×10×1 µm; A_p → B at 1 s^{-1}, B → A_p at 0.5 s^{-1}·[B]; B_0 = 0.2 µM; dt = 0.01; 2 s; 8 seeds). `to_sbml` writes it to SBML Spatial (A particle, B continuous); it is then run four ways: - native from the original model (`to_biomodel` VCML) and native from the SBML (VCell's SBML import → libvcell → vcell-fvsolver); - co-sim from the Python `HybridModel` and co-sim from `from_sbml`.

OBSERVABLE `b6_model_round_trip_exact`, `b6_cosim_vs_cosim_sbml_identical`, `b6_native_vs_native_sbml_identical`, `b6_cosim_sbml_vs_native_sbml_a_max_z`, `b6_cosim_sbml_vs_native_sbml_b_max_z`…

Model & key assumptions

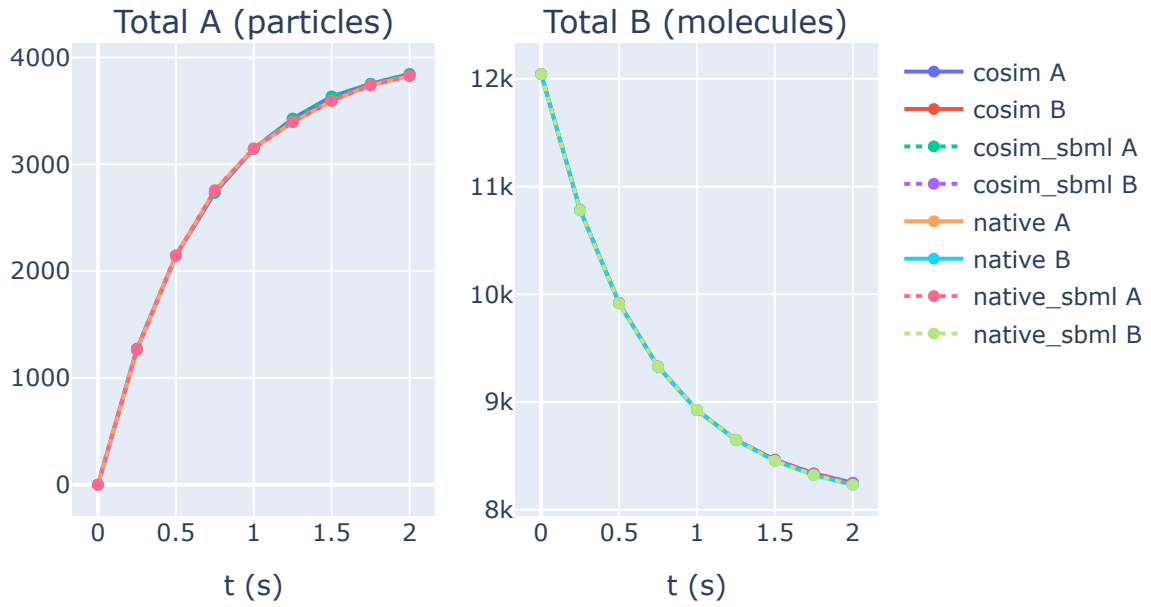
Methods · Simulations

Model	`two_way_exchange`
Sims	1 run
Tests	✓✓✓ 3/3 passed

Visualizations

SBML SPATIAL HYBRID

B6: the A3 two-way exchange from SBML Spatial (dotted) and from th



Finding

F-01 · confirms ● Observed

The hybrid model round-trips exactly through SBML Spatial: from_sbml(to_sbml(model)) equals the original HybridModel (b6_model_round_trip_exact = 1.0), including each species' continuous/particle representation and all mass-action rate constants.

from test: model-round-trips-exactly == 1

runs: `default`

F-02 · confirms ● Observed

Each solver reproduces itself from the SBML seed for seed: the co-sim from SBML is bit-identical to the co-sim from the Python model (b6_cosim_vs_cosim_sbml_identical = 1.0, max |z| = 0) and the native run from SBML is bit-identical to the native run from the original VCML (b6_native_vs_native_sbml_identical = 1.0).

from test: native-from-sbml-is-identical == 1

runs: `default`

F-03 · confirms ● Observed

Across solvers the two SBML runs agree to within sampling: max two-sample |z| is 1.45 for A (b6_cosim_sbml_vs_native_sbml_a_max_z) and 1.14 for B, with native A ending at 3826.6 vs the co-sim's 3845.0, and native conserves A + B to +0.12% (b6_native_mass_balance_final_rel = 0.0012). The two are not bit-identical (b6_cosim_sbml_vs_native_sbml_identical = 0.0), as expected for two independent

solvers of the same SBML. Reaching this needed VCell's SBML import to convert the hybrid to VCell's default units, since its spatial stochastic math is only consistent there.

from test: native-from-sbml-is-identical == 1

runs: `default`

Verdict

● PASSED

Gate cleared · Accepted

Results (results/metrics.json, 2026-10-08; released stack: vcell-fvsolver 0.11.1, libvcell 0.0.19, pyvcell 0.4.2): - **The model round-trips exactly** through SBML Spatial, including the representations and all rate constants. - **Each solver reproduces itself from the SBML, seed for seed:** co-sim and native runs from the SBML are bit-identical to the runs from the original model. - **Across solvers (rerun 2026-10-08 on the released vcell-fvsolver 0.11.1, which carries 32-#34): native and co-sim agree (max |z| 1.45 for A, 1.14 for B; native A ends at 3827 vs** 3845) and native conserves A + B to +0.12%. On 0.10.7, native over-created at boundary nodes (virtualcell/vcell-fvsolver 24): **A ended at 2.3× the co-sim's (|z| 135) and A + B grew 76%.**

▾ Technical record — model internals, full figure set, tests, provenance & follow-ups

Model — composite internals



Static bigraph of `viva_pde_particle.composites.examples.two_way_exchange` — 0 processes wired to 4 compartments. ● writes · ● reads.

`viva_pde_particle.composites.examples.two_way_exchange`

Simulation run

`default`

`model-round-trips-exactly` authored: PASS

`cosim-from-sbml-is-identical` authored: PASS

`native-from-sbml-is-identical` authored: PASS

Reproduce: `vwb run study sbml-spatial-hybrid`
sbml-roundtrip-and-run: `vwb run study sbml-spatial-hybrid --variant sbml-roundtrip-and-run`

Tests

from_sbml(to_sbml(model)) equals the original HybridModel (grid, species, representations, reactions). == 1 pending

`model-round-trips-exactly · primary · derived_scalar`

The SBML carries every model quantity; rate constants are recovered exactly from VCell's mass-action rate.

The co-simulation from the SBML reproduces the co-simulation from the original model seed for seed. == 1 pending

`cosim-from-sbml-is-identical · primary · derived_scalar`

Same HybridModel, same seeds.

The native VCell run from the SBML reproduces the native run from the original VCML seed for seed. == 1 pending

`native-from-sbml-is-identical · primary · derived_scalar`

VCell's SBML import yields math equivalent to the original's (SBMLSpatialHybridTest), so the solver input is the same.

FEEDBACK · THIS WHOLE STUDY 👍 Looks right ✏️ Needs revision

Add a note...

Synthesis & gaps

Where results agree, conflict, or leave open claims — and the priority follow-ups that would resolve them.

Contradictions & uncertainty

> FINDING CONFLICT **— Same ball, two discretizations: native VCell (Cartesian + membrane numerics) vs co-simulation (body-fitted P1)**
>
> The exchange case diverges and fails its band: total A differs at max two-sample |z| = 12.1 (b2b_exchange_a_totals_max_z, band < 4.0), with native A ending 8.1% below the co-sim (b2b_exchange_native_minus_cosim_a_final_rel = −0.0808) and native signed mass balance at −6.8% (b2b_native_exchange_mass_

Same ball, two discretizations: native VCell (Cartesian + membrane numerics) vs co-simulation (body-fitted P1) SUPPORTED

FINDING CONFLICT — **Fields next to a curved membrane: native VCell on pyvcell's smoothed grid vs co-sim on a body-fitted mesh**

The native outer shell (r > 3.5 µm) sits 2.3% below the interior (b2c_native_shell_bias = −0.0227, z = −20.1), failing the ±1% unbiasedness band, while the co-sim shell is unbiased at +0.12% (b2c_cosim_shell_bias = 0.0012, z = 1.17).

Fields next to a curved membrane: native VCell on pyvcell's smoothed grid vs co-sim on a body-fitted mesh SUPPORTED

FINDING CONFLICT — **Removing the co-simulation's boundary artifact: particle transfer, membrane geometry and curved-compartment creation**

The mesh-boundary membrane removes the outer-shell bias: with grid transfer and the mesh membrane the shell bias is +0.12% (z = 1.2), inside the ±0.5% band, down from +1.27% (z = 10.3) with the exact-sphere membrane.

Removing the co-simulation's boundary artifact: particle transfer, membrane geometry and curved-compartment creation SUPPORTED

FINDING CONFLICT — **Removing the co-simulation's boundary artifact: particle transfer, membrane geometry and curved-compartment creation**

The hypothesis was wrong: the membrane, not the transfer, drives the bias.

Removing the co-simulation's boundary artifact: particle transfer, membrane geometry and curved-compartment creation SUPPORTED

FINDING CONFLICT — **The co-simulation on VCell's own geometry: staircase PDE, smooth membrane, accessible-volume correction**

The adapters+volumes correction makes the coupling unbiased on VCell's own geometry: the outer-shell bias is +0.017% (b2e_adapters_volumes_shell_bias = 0.000167, within ±0.5%) and the interior is +0.20% of E[B] (b2e_adapters_volumes_interior_rel = 0.00203, within ±1%).

The co-simulation on VCell's own geometry: staircase PDE, smooth membrane, accessible-volume correction SUPPORTED

FINDING CONFLICT — **The co-simulation on VCell's own geometry: staircase PDE, smooth membrane, accessible-volume correction**

The intentional regression check FAILS as designed: with correction="none" (full voxel volumes) the outer-shell bias is −1.36% (b2e_none_shell_bias = −0.0136, z = −12.5), outside the ±0.5% band.

The co-simulation on VCell's own geometry: staircase PDE, smooth membrane, accessible-volume correction SUPPORTED

FINDING CONFLICT — **One non-convex GeometryDescription, three solver paths: native VCell, co-sim on VCell's geometry, co-sim on a Netgen mesh**

The accessible-volume correction keeps the co-sim conservative and unbiased on exactly VCell's thin-necked geometry: exchange A+B balance is -0.07% (±1.5% band) and the near-membrane bias is -0.02% (z = -1.2, ±0.5% band), with the steady ratio off by 0.39%.

One non-convex GeometryDescription, three solver paths: native VCell, co-sim on VCell's geometry, co-sim on a Netgen mesh SUPPORTED

FINDING CONFLICT — **One non-convex GeometryDescription, three solver paths: native VCell, co-sim on VCell's geometry, co-sim on a Netgen mesh**

The body-fitted Netgen mesh path is also conservative: exchange balance is -0.12% (±1.5% band) with a +0.04% near-membrane bias, against a -3.0% mesh cell-volume error.

One non-convex GeometryDescription, three solver paths: native VCell, co-sim on VCell's geometry, co-sim on a Netgen mesh SUPPORTED

FINDING CONFLICT

— One non-convex GeometryDescription, three solver paths: native VCell, co-sim on VCell's geometry, co-sim on a Netgen mesh

On a thin-necked shape the uncorrected smooth-vs-staircase mismatch is severe and the native path fails the band: native VCell loses -25.9% of A+B in exchange and misses the steady ratio by 8.9%, driven by the -12.4% smooth-membrane volume error that shrinks the neck and small lobes.

One non-convex GeometryDescription, three solver paths: native VCell, co-sim on VCell's geometry, co-sim on a Netgen mesh SUPPORTED

FINDING CONFLICT

— Operator-splitting schemes for PDE/particle coupling: Jacobi (fvsolver), Gauss–Seidel, Strang

Strang's error stays within ~2× of the sampling floor across the whole sweep (0.000405 at k=2 to 0.000859 at k=32), so its fitted order (0.34) is floor-dominated and not resolvable — more seeds or larger τ would be needed to measure the expected second order.

Operator-splitting schemes for PDE/particle coupling: Jacobi (fvsolver), Gauss–Seidel, Strang SUPPORTED

CAUSAL CONTINUITY

These studies take their inputs from outside the chain (a supplied feed or reset state) rather than from an upstream study's output — so their result does not yet rest on a modeled upstream mechanism:

PDE engine swap, FV (VCell FV_SOLVER scheme) vs FEniCSx Q1 under the same coupling · Operator-splitting schemes for PDE/particle coupling: Jacobi (fvsolver), Gauss–Seidel, Strang · The hybrid model described in SBML Spatial, run on both solvers

Validation & reproducibility

Automated, code-computed consistency checks (no AI): conservation/invariant ledgers, computed-vs-authored verdicts, run reproducibility (environment + determinism), parameter drift and acceptance-criterion coverage. Each study's own ledger predicates (e.g. mass/charge balance) are its gating behavior tests, listed on the study's card (its Tests).

Diagnostic study needed `cosim-boundary-transfer`

Study 'cosim-boundary-transfer' verdict is 'failed' with no diagnostic study seeded

Diagnostic study needed `dumbbell-three-ways`

Study 'dumbbell-three-ways' verdict is 'failed' with no diagnostic study seeded

Diagnostic study needed `near-membrane-fields`

Study 'near-membrane-fields' verdict is 'failed' with no diagnostic study seeded

Diagnostic study needed `sphere-cosim-vs-native`

Study 'sphere-cosim-vs-native' verdict is 'failed' with no diagnostic study seeded

Diagnostic study needed `vcell-geometry-cosim`

Study 'vcell-geometry-cosim' verdict is 'failed' with no diagnostic study seeded

Stale finding `F-01`

Finding 'F-01' declared a next_action but no study was seeded

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Stale finding `F-03`
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Finding 'F-03' declared a next_action but no study was seeded

Stale finding F-03

Finding 'F-03' declared a next_action but no study was seeded

▼ REPORT READINESS — 122 FINDINGS ACROSS 11 CHECKS (0 ERRORS, 122 WARNINGS)

44 band test missing cites warning

`cosim-boundary-transfer, dumbbell-three-ways, fenicsx-same-problem, handoff-units-and-binning +8`

18 runs without emitter warning

`bimolecular-hybrid, cosim-boundary-transfer, coupled-sparks-fast-diffusion, coupling-interval-convergence +14`

18 unresolved composite warning

`bimolecular-hybrid, cosim-boundary-transfer, coupled-sparks-fast-diffusion, coupling-interval-convergence +14`

14 dag edges legacy only warning

`bimolecular-hybrid, cosim-boundary-transfer, coupled-sparks-fast-diffusion, coupling-interval-convergence +10`

6 authored confidence diverges warning

`cosim-boundary-transfer, dumbbell-three-ways, near-membrane-fields, single-channel-fokker-planck +2`

6 investigation narrative spine required warning

`cosim-vs-embedded-hybrid, hybrid-generalizability, vcell-hybrid-paper-benchmarks`

5 baseline config carry warning

`cosim-boundary-transfer, dumbbell-three-ways, near-membrane-fields, single-channel-fokker-planck +1`

5 reviewer clarity ambiguity warning

`cosim-boundary-transfer, dumbbell-three-ways, near-membrane-fields, sphere-cosim-vs-native +1`

3 investigation missing adversarial warning

`cosim-vs-embedded-hybrid, hybrid-generalizability, vcell-hybrid-paper-benchmarks`

2 unearned unit labels warning

`near-membrane-fields, sphere-cosim-vs-native`

1 dag edges legacy redundant warning

`field-modulated-decay`

Evidence register

Every study in the campaign (9) — scientific verdict, run status, and one-line result, each linked to its full study card above.

STUDY	Verdict	Run	Result
PDE engine swap, FV (VCell FV_SOLVER scheme) vs FEniCSx Q1 under the same coupling	SUPPORTED	ran	All three PDE engines are second order on the cosine mode: fitted h-orders are 1.97 (FV), 1.98 (Q1 lumped) and 2.04 (Q1 consistent), all inside the [1.8, 2.2] band, with L∞ errors at N=81 of 5.5e-6 (FV), 1.6e-5 (lumped)
Hybrid co-simulation with a body-fitted unstructured PDE mesh (ball)	SUPPORTED	ran	The particle reaction is geometry-independent: on the body-fitted tetrahedral ball mesh the mean A follows $A_0 \cdot e^{(-kt)}$ with a max |z| of 1.54 over time (well under the 4.0 band), so the unstructured domain does not pertur
Same ball, two discretizations: native VCell (Cartesian + membrane numerics) vs co-simulation (body-fitted P1)	SUPPORTED	ran	At this resolution VCell's Cartesian representation of the ball is the more accurate geometry: −0.47% volume error (b2b_native_staircase_volume_rel_err = −0.0047) against −1.4% for the co-sim body-fitted mesh at h = 0.8
Fields next to a curved membrane: native VCell on pyvcell's smoothed grid vs co-sim on a body-fitted mesh	SUPPORTED	ran	The node-centred smoothed cell domain reproduces the ball to within −0.50% (b2c_node_centred_smoothed_volume_rel_err = −0.0050), against −8.4% for the pyvcell export taken as-is, and smoothing cuts the RMS radial error o
Removing the co-simulation's boundary artifact: particle transfer, membrane geometry and curved-compartment creation	SUPPORTED	ran	The mesh-boundary membrane removes the outer-shell bias: with grid transfer and the mesh membrane the shell bias is +0.12% (z = 1.2), inside the ±0.5% band, down from +1.27% (z = 10.3) with the exact-sphere membrane.
The co-simulation on VCell's own geometry: staircase PDE, smooth membrane, accessible-volume correction	SUPPORTED	ran	The adapters+volumes correction makes the coupling unbiased on VCell's own geometry: the outer-shell bias is +0.017% (b2e_adapters_volumes_shell_bias = 0.000167, within ±0.5%) and the interior is +0.20% of E[B] (b2e_adap
One non-convex GeometryDescription, three solver paths: native VCell, co-sim on VCell's geometry, co-sim on a Netgen mesh	SUPPORTED	ran	The accessible-volume correction keeps the co-sim conservative and unbiased on exactly VCell's thin-necked geometry: exchange A+B balance is -0.07% (±1.5% band) and the near-membrane bias is -0.02% (z = -1.2, ±0.5% band)
Operator-splitting schemes for PDE/particle coupling: Jacobi (fvsolver), Gauss–Seidel, Strang	SUPPORTED	ran	All three lagged splittings are first order in the coupling interval τ: the fitted order is 1.23 for Jacobi (fvsolver), 1.01 for Gauss–Seidel (particles-first) and 1.08 for Gauss–Seidel (pde-first).

STUDY	Verdict	Run	Result
The hybrid model described in SBML Spatial, run on both solvers	SUPPORTED	ran	The hybrid model round-trips exactly through SBML Spatial: from_sbml(to_sbml(model)) equals the original HybridModel (b6_model_round_trip_exact = 1.0), including each species' continuous/particle representation and all m

Leave an overall note, and per-study notes on each study card. Then **Review & export** assembles them into a YAML feedback report.

FEEDBACK · THE WHOLE REPORT 👍 Looks right ✏️ Needs revision

Add a note…

Rendered from workspace `viva-pde-particle` · deterministic from investigation + study YAML.

`investigations/hybrid-generalizability/investigation.yaml · studies/*/study.yaml · loop-trajectory JSON`